# Tables of Large Sets LS$[N](t, k, v)$ of Prime Size for $2 \le t \le 6$ and $v \le 100$

Supplement to "New Large Sets of $t$-Designs from Affine Groups"

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## Scope and conventions

The tables cover $N \in \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43\}$, $2 \le t \le 6$, $t < k \le v/2$ and $v \le 100$. A parameter set $(N; t, k, v)$ is *admissible* when $N$ divides $\binom{v-i}{k-i}$ for $0 \le i \le t$. For each $N$ and $t$ the table has one row for each $v$ with an admissible $k$. The columns list the values of $k$ for which a large set exists, the values for which existence follows from the results of the accompanying paper (in bold), the values for which a large set is known not to exist, and the admissible values that are open. Parameter sets with $v - k < t$ are omitted.

Every entry marked as existing is derived by a program from a list of base results by the recursive theorems listed below. The file `large_sets_table.csv` gives a justification for each of the 50441 admissible parameter sets: its source, or the rule and the parameter sets from which it was derived.

## Recursive theorems used

1. All admissible large sets of 1-designs exist (Baranyai, Hartman).
2. An LS$[N](t, k, v)$ gives an LS$[N](t - i, k - j, v - l)$ for $0 \le j \le l \le i \le t$, and an LS$[N](t, v - k, v)$.
3. An LS$[N](t, k, v)$ and an LS$[N](t, k + 1, v)$ give an LS$[N](t, k + 1, v + 1)$ (Ajoodani-Namini and Khosrovshahi).
4. If LS$[N](t, i, v)$ exist for $t < i \le k$ and an LS$[N](t, k, u)$ exists, then LS$[N](t, k, u + l(v - t))$ exist for $l \ge 1$ (Ajoodani-Namini and Khosrovshahi).
5. For a prime $p$, an LS$[p](t, k, v - 1)$ gives LS$[p](t + 1, pk + i, pv + j)$ for $0 \le j < i \le p - 1$ (Ajoodani-Namini) and LS$[p](t, pk + i, pv + j)$ for $-p \le j < i \le p - 1$ (Ajoodani-Namini, Tayfeh-Rezaie).
6. An LS$[N](t, k, 2k + 1)$ with $t$ even, or with $N = 2$, gives an LS$[N](t + 1, k + 1, 2k + 2)$ (Alltop).
7. For a prime $p$: if LS$[p](t, i, p^f + t)$ exist for $t < i \le \min(k, (p^f + t)/2)$ and $t \le v \bmod p^f < k \bmod p^f$, then an LS$[p](t, k, v)$ exists (Khosrovshahi and Tayfeh-Rezaie).

Non-existence is propagated through item 2.

## Base results

General theorems: $t = 2$, $k = 3$, all $N$, $v \ne 7$ (Lu, Teirlinck). $N = 2$: $t = 2$ and all $k$ (Ajoodani-Namini), $3 \le t \le 5$ and $k \le 15$, $t = 6$ and $k \in \{7, 8, 9\}$. $N = 3$: $t \in \{2, 3\}$ and $k \le 80$, $t = 4$ and $k \le 10$. $N = 5$: $t = 2$, $k \le 8$, and $t = 3$, $k \le 6$. $N = 7$: $t \in \{2, 3\}$, $k \le 6$. $N = 11$: $t = 2$, $k \le 10$. $N \in \{13, 17, 19, 23\}$: $t = 2$, $k \in \{3, 4\}$. $N = 29$: $t = 2$, $k \le 28$. Minimum-index large sets of 3-$(v, 4, \lambda)$ designs for $v \equiv 0 \pmod 3$ (Teirlinck).

Directly constructed large sets, with the keys used in the CSV file: KMS91 (Kramer, Magliveras, Stinson, Australas. J. Combin. 3 (1991) 191–205), Chouinard83 (Discrete Math. 45 (1983) 297–300), KMO93 (Kramer, Magliveras, O'Brien, Australas. J. Combin. 7 (1993) 189–193), CCFK90 (Chee, Colbourn, Furino, Kreher, Australas. J. Combin. 2 (1990) 111–119), CM98 (Chee, Magliveras, J. Combin. Des. 6 (1998) 293–308), Denniston83 (Ann. Discrete Math. 18 (1983) 291–294), Eslami03 (J. Combin. Des. 11 (2003) 312–316), CuMa99 (Cusack, Magliveras, Des. Codes Cryptogr. 18 (1999) 81–87), LMW01 (Laue, Magliveras, Wassermann, J. Combin. Des. 9 (2001) 40–59, including the large sets obtained from its Theorem 5 for affine and projective groups with orbits of two lengths), LOTW07 (Laue, Omidi, Tayfeh-Rezaie, Wassermann, J. Combin. Des. 15 (2007) 210–220, Tables II, III and IV), KT04 (Khosrovshahi, Tayfeh-Rezaie, lecture notes, 2004), Teirlinck84 (Ars Combin. 17

(1984) 173–176), Teirlinck89 (Discrete Math. 77 (1989) 345–356), and the semiregular families from ASL$(1,q)$ and AGL$(1,2^n)$ of the accompanying paper.

Non-existence: LS[5](2, 3, 7) (Cayley), LS[7](3, 4, 10) (Kramer and Mesner), LS[11](4, 5, 15) (Mendelsohn and Hung), LS[13](4, 5, 17) (Östergård and Pottonen).

## Limitations

The tables contain only what follows from the sources above. Results published after 2010 are not included. For $N = 2$ no results beyond $k = 15$ for $t \geq 3$ are included. Large sets obtained by combining orbits of PSL$(2,q)$ and PGL$(2,q)$ are included for $q < 60$ from LOTW07 and for larger $q$ only when the orbits have at most two lengths. One entry of Table II of LOTW07, $N = 29$ for $(3, 26, 60)$, is not admissible and was omitted. An entry marked open may therefore be settled in the literature.

## Summary

Number of existing parameter sets over number of admissible ones.

| $N$ | $t=2$ | $t=3$ | $t=4$ | $t=5$ | $t=6$ |
|---|---|---|---|---|---|
| 2 | 1456/1456 | 1231/1287 | 1052/1155 | 878/1033 | 370/951 |
| 3 | 1299/1299 | 1164/1164 | 605/1039 | 166/928 | 28/827 |
| 5 | 631/922 | 304/756 | 26/640 | 2/580 | 0/522 |
| 7 | 859/916 | 350/721 | 48/602 | 6/517 | 0/462 |
| 11 | 656/656 | 69/512 | 0/384 | 0/276 | 0/184 |
| 13 | 286/769 | 52/629 | 6/501 | 3/389 | 1/289 |
| 17 | 304/798 | 61/693 | 0/594 | 0/504 | 0/420 |
| 19 | 488/834 | 87/714 | 0/600 | 0/495 | 0/396 |
| 23 | 244/878 | 81/778 | 0/682 | 0/592 | 0/506 |
| 29 | 905/905 | 124/827 | 1/751 | 0/679 | 0/609 |
| 31 | 268/847 | 107/763 | 0/681 | 0/603 | 0/527 |
| 37 | 475/856 | 16/805 | 0/755 | 0/707 | 0/660 |
| 41 | 257/890 | 74/833 | 0/777 | 0/723 | 0/670 |
| 43 | 287/862 | 60/802 | 0/743 | 0/686 | 0/630 |

## Tables

### LS[2](2, $k$, $v$)

All 1456 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ are realised.

### LS[2](3, $k$, $v$)

1231 of 1287 admissible parameter sets exist, 56 are open.

| $v$ | exists, $k=$ | new | none | open, $k=$ |
|---|---|---|---|---|
| 11 | 4–5 | | | |
| 12 | 5–6 | | | |
| 13 | 6 | | | |
| 14 | 7 | | | |
| 19 | 4–9 | | | |
| 20 | 5–10 | | | |
| 21 | 6–10 | | | |
| 22 | 7–11 | | | |
| 23 | 8–11 | | | |
| 24 | 9–12 | | | |
| 25 | 10–12 | | | |
| 26 | 11–13 | | | |
| 27 | 4–7, 12–13 | | | |
| 28 | 5–7, 13–14 | | | |
| 29 | 6–7, 14 | | | |
| 30 | 7, 15 | | | |
| 35 | 4–15 | | | 16–17 |
| 36 | 5–15, 18 | | | 16–17 |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 37 | 6–15 | | | 16–18 |
| 38 | 7–19 | | | |
| 39 | 8–19 | | | |
| 40 | 9–20 | | | |
| 41 | 10–20 | | | |
| 42 | 11–21 | | | |
| 43 | 4–7, 12–21 | | | |
| 44 | 5–7, 13–22 | | | |
| 45 | 6–7, 14–22 | | | |
| 46 | 7, 15–23 | | | |
| 47 | 16–23 | | | |
| 48 | 17–24 | | | |
| 49 | 18–24 | | | |
| 50 | 19–25 | | | |
| 51 | 4–15, 20–25 | | | |
| 52 | 5–15, 21–26 | | | |
| 53 | 6–15, 22–26 | | | |
| 54 | 7–15, 23–27 | | | |
| 55 | 8–15, 24–27 | | | |
| 56 | 9–15, 25–28 | | | |
| 57 | 10–15, 26–28 | | | |
| 58 | 11–15, 27–29 | | | |
| 59 | 4–7, 12–15, 20–23, 28–29 | | | |
| 60 | 5–7, 13–15, 21–23, 29–30 | | | |
| 61 | 6–7, 14–15, 22–23, 30 | | | |
| 62 | 7, 15, 23, 31 | | | |
| 67 | 4–15, 20–31 | | | 16–19, 32–33 |
| 68 | 5–15, 19, 21–31, 34 | | | 16–18, 20, 32–33 |
| 69 | 6–15, 22–31 | | | 16–21, 32–34 |
| 70 | 7–31, 33, 35 | | | 32, 34 |
| 71 | 8–31 | | | 32–35 |
| 72 | 9–31, 33, 35–36 | | | 32, 34 |
| 73 | 10–31, 36 | | | 32–35 |
| 74 | 11–31, 33, 35, 37 | | | 32, 34, 36 |
| 75 | 4–7, 12–31, 36–37 | | | 32–35 |
| 76 | 5–7, 13–38 | | | |
| 77 | 6–7, 14–38 | | | |
| 78 | 7, 15–39 | | | |
| 79 | 16–39 | | | |
| 80 | 17–40 | | | |
| 81 | 18–40 | | | |
| 82 | 19–41 | | | |
| 83 | 4–15, 20–41 | | | |
| 84 | 5–15, 21–42 | | | |
| 85 | 6–15, 22–42 | | | |
| 86 | 7–15, 23–43 | | | |
| 87 | 8–15, 24–43 | | | |
| 88 | 9–15, 25–44 | | | |
| 89 | 10–15, 26–44 | | | |
| 90 | 11–15, 27–45 | | | |
| 91 | 4–7, 12–15, 20–23, 28–45 | | | |
| 92 | 5–7, 13–15, 21–23, 29–46 | | | |
| 93 | 6–7, 14–15, 22–23, 30–46 | | | |
| 94 | 7, 15, 23, 31–47 | | | |
| 95 | 32–47 | | | |
| 96 | 33–48 | | | |
| 97 | 34–48 | | | |
| 98 | 35–49 | | | |
| 99 | 4–15, 20–31, 36–49 | | | 16–19 |
| 100 | 5–15, 21–31, 37–50 | | | 16–20 |

## LS[2](4, $k$, $v$)

1052 of 1155 admissible parameter sets exist, 103 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 12 | 5–6 | | | |
| 13 | 6 | | | |
| 14 | 7 | | | |
| 20 | 5–10 | | | |
| 21 | 6–10 | | | |
| 22 | 7–11 | | | |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 23 | 8–11 | | | |
| 24 | 9–12 | | | |
| 25 | 10–12 | | | |
| 26 | 11–13 | | | |
| 27 | 12–13 | | | |
| 28 | 5–7, 13–14 | | | |
| 29 | 6–7, 14 | | | |
| 30 | 7, 15 | | | |
| 36 | 5–15 | | | 16–18 |
| 37 | 6–15 | | | 16–18 |
| 38 | 7–15 | | | 16–19 |
| 39 | 8–15 | | | 16–19 |
| 40 | 9–20 | | | |
| 41 | 10–20 | | | |
| 42 | 11–21 | | | |
| 43 | 12–21 | | | |
| 44 | 5–7, 13–22 | | | |
| 45 | 6–7, 14–22 | | | |
| 46 | 7, 15–23 | | | |
| 47 | 16–23 | | | |
| 48 | 17–24 | | | |
| 49 | 18–24 | | | |
| 50 | 19–25 | | | |
| 51 | 20–25 | | | |
| 52 | 5–15, 21–26 | | | |
| 53 | 6–15, 22–26 | | | |
| 54 | 7–15, 23–27 | | | |
| 55 | 8–15, 24–27 | | | |
| 56 | 9–15, 25–28 | | | |
| 57 | 10–15, 26–28 | | | |
| 58 | 11–15, 27–29 | | | |
| 59 | 12–15, 28–29 | | | |
| 60 | 5–7, 13–15, 21–23, 29–30 | | | |
| 61 | 6–7, 14–15, 22–23, 30 | | | |
| 62 | 7, 15, 23, 31 | | | |
| 68 | 5–15, 21–31 | | | 16–20, 32–34 |
| 69 | 6–15, 22–31 | | | 16–21, 32–34 |
| 70 | 7–15, 23–31 | | | 16–22, 32–35 |
| 71 | 8–15, 24–31 | | | 16–23, 32–35 |
| 72 | 9–31 | | | 32–36 |
| 73 | 10–31 | | | 32–36 |
| 74 | 11–31, 37 | | | 32–36 |
| 75 | 12–31 | | | 32–37 |
| 76 | 5–7, 13–31, 37–38 | | | 32–36 |
| 77 | 6–7, 14–31, 38 | | | 32–37 |
| 78 | 7, 15–31, 33, 35, 37, 39 | | | 32, 34, 36, 38 |
| 79 | 16–31 | | | 32–39 |
| 80 | 17–40 | | | |
| 81 | 18–40 | | | |
| 82 | 19–41 | | | |
| 83 | 20–41 | | | |
| 84 | 5–15, 21–42 | | | |
| 85 | 6–15, 22–42 | | | |
| 86 | 7–15, 23–43 | | | |
| 87 | 8–15, 24–43 | | | |
| 88 | 9–15, 25–44 | | | |
| 89 | 10–15, 26–44 | | | |
| 90 | 11–15, 27–45 | | | |
| 91 | 12–15, 28–45 | | | |
| 92 | 5–7, 13–15, 21–23, 29–46 | | | |
| 93 | 6–7, 14–15, 22–23, 30–46 | | | |
| 94 | 7, 15, 23, 31–47 | | | |
| 95 | 32–47 | | | |
| 96 | 33–48 | | | |
| 97 | 34–48 | | | |
| 98 | 35–49 | | | |
| 99 | 36–49 | | | |
| 100 | 5–15, 21–31, 37–50 | | | 16–20 |

## LS[2](5, $k$, $v$)

878 of 1033 admissible parameter sets exist, 155 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 13 | 6 | | | |
| 14 | 7 | | | |
| 21 | 6–10 | | | |
| 22 | 7–11 | | | |
| 23 | 8–11 | | | |
| 24 | 9–12 | | | |
| 25 | 10–12 | | | |
| 26 | 11–13 | | | |
| 27 | 12–13 | | | |
| 28 | 13–14 | | | |
| 29 | 6–7, 14 | | | |
| 30 | 7, 15 | | | |
| 37 | 6–15 | | | 16–18 |
| 38 | 7–15 | | | 16–19 |
| 39 | 8–15 | | | 16–19 |
| 40 | 9–15 | | | 16–20 |
| 41 | 10–15 | | | 16–20 |
| 42 | 11–21 | | | |
| 43 | 12–21 | | | |
| 44 | 13–22 | | | |
| 45 | 6–7, 14–22 | | | |
| 46 | 7, 15–23 | | | |
| 47 | 16–23 | | | |
| 48 | 17–24 | | | |
| 49 | 18–24 | | | |
| 50 | 19–25 | | | |
| 51 | 20–25 | | | |
| 52 | 21–26 | | | |
| 53 | 6–15, 22–26 | | | |
| 54 | 7–15, 23–27 | | | |
| 55 | 8–15, 24–27 | | | |
| 56 | 9–15, 25–28 | | | |
| 57 | 10–15, 26–28 | | | |
| 58 | 11–15, 27–29 | | | |
| 59 | 12–15, 28–29 | | | |
| 60 | 13–15, 29–30 | | | |
| 61 | 6–7, 14–15, 22–23, 30 | | | |
| 62 | 7, 15, 23, 31 | | | |
| 69 | 6–15, 22–31 | | | 16–21, 32–34 |
| 70 | 7–15, 23–31 | | | 16–22, 32–35 |
| 71 | 8–15, 24–31 | | | 16–23, 32–35 |
| 72 | 9–15, 25–31 | | | 16–24, 32–36 |
| 73 | 10–15, 26–31 | | | 16–25, 32–36 |
| 74 | 11–31 | | | 32–37 |
| 75 | 12–31 | | | 32–37 |
| 76 | 13–31 | | | 32–38 |
| 77 | 6–7, 14–31, 38 | | | 32–37 |
| 78 | 7, 15–31, 39 | | | 32–38 |
| 79 | 16–31 | | | 32–39 |
| 80 | 17–31 | | | 32–40 |
| 81 | 18–31 | | | 32–40 |
| 82 | 19–31, 33, 35, 37, 39, 41 | | | 32, 34, 36, 38, 40 |
| 83 | 20–31 | | | 32–41 |
| 84 | 21–42 | | | |
| 85 | 6–15, 22–42 | | | |
| 86 | 7–15, 23–43 | | | |
| 87 | 8–15, 24–43 | | | |
| 88 | 9–15, 25–44 | | | |
| 89 | 10–15, 26–44 | | | |
| 90 | 11–15, 27–45 | | | |
| 91 | 12–15, 28–45 | | | |
| 92 | 13–15, 29–46 | | | |
| 93 | 6–7, 14–15, 22–23, 30–46 | | | |
| 94 | 7, 15, 23, 31–47 | | | |
| 95 | 32–47 | | | |
| 96 | 33–48 | | | |
| 97 | 34–48 | | | |
| 98 | 35–49 | | | |
| 99 | 36–49 | | | |
| 100 | 37–50 | | | |

## LS[2](6, $k$, $v$)

370 of 951 admissible parameter sets exist, 581 are open.

| $v$ | exists, $k$ = | new | none | open, $k$ = |
|---|---|---|---|---|
| 14 | 7 | | | |
| 22 | 7–9, 11 | | | 10 |
| 23 | 8–9 | | | 10–11 |
| 24 | 9, 12 | | | 10–11 |
| 25 | | | | 10–12 |
| 26 | 13 | | | 11–12 |
| 27 | | | | 12–13 |
| 28 | 13–14 | | | |
| 29 | 14 | | | |
| 30 | 7, 15 | | | |
| 38 | 7–9, 15 | | | 10–14, 16–19 |
| 39 | 8–9 | | | 10–19 |
| 40 | 9 | | | 10–20 |
| 41 | | | | 10–20 |
| 42 | | | | 11–21 |
| 43 | | | | 12–21 |
| 44 | 13–19, 21–22 | | | 20 |
| 45 | 14–19, 22 | | | 20–21 |
| 46 | 7, 15–19, 21, 23 | | | 20, 22 |
| 47 | 16–19 | | | 20–23 |
| 48 | 17–19, 21, 23–24 | | | 20, 22 |
| 49 | 18–19, 24 | | | 20–23 |
| 50 | 19, 21, 23, 25 | | | 20, 22, 24 |
| 51 | | | | 20–25 |
| 52 | 21, 23, 25–26 | | | 22, 24 |
| 53 | 26 | | | 22–25 |
| 54 | 7–9, 15, 23, 25, 27 | | | 10–14, 24, 26 |
| 55 | 8–9 | | | 10–15, 24–27 |
| 56 | 9, 25–28 | | | 10–15 |
| 57 | 26–28 | | | 10–15 |
| 58 | 27–29 | | | 11–15 |
| 59 | 28–29 | | | 12–15 |
| 60 | 13–15, 29–30 | | | |
| 61 | 14–15, 30 | | | |
| 62 | 7, 15, 23, 31 | | | |
| 70 | 7–9, 15, 23, 31 | | | 10–14, 16–22, 24–30, 32–35 |
| 71 | 8–9 | | | 10–35 |
| 72 | 9 | | | 10–36 |
| 73 | | | | 10–36 |
| 74 | | | | 11–37 |
| 75 | | | | 12–37 |
| 76 | 13–19, 21, 23, 25, 27, 29–31 | | | 20, 22, 24, 26, 28, 32–38 |
| 77 | 14–19, 30–31 | | | 20–29, 32–38 |
| 78 | 7, 15–19, 21, 23, 25, 27, 29, 31, 39 | | | 20, 22, 24, 26, 28, 30, 32–38 |
| 79 | 16–19 | | | 20–39 |
| 80 | 17–19, 21, 23, 25, 27, 29, 31 | | | 20, 22, 24, 26, 28, 30, 32–40 |
| 81 | 18–19 | | | 20–40 |
| 82 | 19, 21, 23, 25, 27, 29, 31 | | | 20, 22, 24, 26, 28, 30, 32–41 |
| 83 | | | | 20–41 |
| 84 | 21, 23, 25, 27, 29, 31 | | | 22, 24, 26, 28, 30, 32–42 |
| 85 | | | | 22–42 |
| 86 | 7–9, 15, 23, 25, 27, 29, 31, 33, 35, 37, 39, 41, 43 | | | 10–14, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42 |
| 87 | 8–9 | | | 10–15, 24–43 |
| 88 | 9, 25–39, 41–44 | | | 10–15, 40 |
| 89 | 26–39, 42–44 | | | 10–15, 40–41 |
| 90 | 27–39, 41, 43–45 | | | 11–15, 40, 42 |
| 91 | 28–39, 44–45 | | | 12–15, 40–43 |
| 92 | 13–15, 29–39, 41–43, 45–46 | | | 40, 44 |
| 93 | 14–15, 30–39, 42–43, 46 | | | 40–41, 44–45 |
| 94 | 7, 15, 23, 31–39, 41, 43, 45, 47 | | | 40, 42, 44, 46 |
| 95 | 32–39 | | | 40–47 |
| 96 | 33–39, 41–43, 45–48 | | | 40, 44 |
| 97 | 34–39, 42–43, 46–48 | | | 40–41, 44–45 |
| 98 | 35–39, 41, 43, 45, 47–49 | | | 40, 42, 44, 46 |
| 99 | 36–39, 48–49 | | | 40–47 |
| 100 | 37–39, 41–43, 45–47, 49–50 | | | 40, 44, 48 |

## LS[3](2, $k$, $v$)

All 1299 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ are realised.

## LS[3](3, $k$, $v$)

All 1164 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ are realised.

## LS[3](4, $k$, $v$)

605 of 1039 admissible parameter sets exist, 434 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 13 | 5–6 | | | |
| 14 | 6–7 | | | |
| 15 | 7 | | | |
| 16 | 8 | | | |
| 22 | 5–8 | | | |
| 23 | 6–8 | | | |
| 24 | 7–8 | | | |
| 25 | 8 | | | |
| 31 | 5–10, 14–15 | | | 11–13 |
| 32 | 6–10, 15–16 | | | 11–14 |
| 33 | 7–10, 16 | | | 11–15 |
| 34 | 8–10, 17 | | | 11–16 |
| 35 | 9–10 | | | 11–17 |
| 36 | 10 | | | 11–18 |
| 37 | | | | 11–18 |
| 38 | | | | 12–19 |
| 39 | 13–19 | | | |
| 40 | 5–8, 14–20 | | | |
| 41 | 6–8, 15–20 | | | |
| 42 | 7–8, 16–21 | | | |
| 43 | 8, 17–21 | | | |
| 44 | 18–22 | | | |
| 45 | 19–22 | | | |
| 46 | 20–23 | | | |
| 47 | 21–23 | | | |
| 48 | 22–24 | | | |
| 49 | 5–8, 14–17, 23–24 | | | |
| 50 | 6–8, 15–17, 24–25 | | | |
| 51 | 7–8, 16–17, 25 | | | |
| 52 | 8, 17, 26 | | | |
| 58 | 5–10, 14–17, 23–26 | | | 11–13, 18–22 |
| 59 | 6–10, 15–17, 24–26 | | | 11–14, 18–23 |
| 60 | 7–10, 16–17, 25–26 | | | 11–15, 18–24 |
| 61 | 8–10, 17, 26 | | | 11–16, 18–25 |
| 62 | 9–10 | | | 11–26 |
| 63 | 10 | | | 11–26 |
| 64 | | | | 11–26 |
| 65 | | | | 12–26 |
| 66 | 13–26 | | | |
| 67 | 5–8, 14–26, 32–33 | | | |
| 68 | 6–8, 15–26, 33–34 | | | |
| 69 | 7–8, 16–26, 34 | | | |
| 70 | 8, 17–26, 35 | | | |
| 71 | 18–26 | | | |
| 72 | 19–26 | | | |
| 73 | 20–26 | | | |
| 74 | 21–26 | | | |
| 75 | 22–26 | | | |
| 76 | 5–8, 14–17, 23–26, 32–35 | | | |
| 77 | 6–8, 15–17, 24–26, 33–35 | | | |
| 78 | 7–8, 16–17, 25–26, 34–35 | | | |
| 79 | 8, 17, 26, 35 | | | |
| 85 | 5–10, 14–17, 23–26, 32–35, 41–42 | | | 11–13, 18–22, 27–31, 36–40 |
| 86 | 6–10, 15–17, 24–26, 33–35, 42–43 | | | 11–14, 18–23, 27–32, 36–41 |
| 87 | 7–10, 16–17, 25–26, 34–35, 43 | | | 11–15, 18–24, 27–33, 36–42 |
| 88 | 8–10, 17, 26, 35, 44 | | | 11–16, 18–25, 27–34, 36–43 |
| 89 | 9–10 | | | 11–44 |
| 90 | 10 | | | 11–45 |
| 91 | | | | 11–45 |
| 92 | | | | 12–46 |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 93 | 13–32, 34–35, 37–38, 40–46 | | | 33, 36, 39 |
| 94 | 5–8, 14–35, 38, 41–47 | | | 36–37, 39–40 |
| 95 | 6–8, 15–35, 42–47 | | | 36–41 |
| 96 | 7–8, 16–35, 37–38, 40–41, 43–48 | | | 36, 39, 42 |
| 97 | 8, 17–35, 38, 41, 44–48 | | | 36–37, 39–40, 42–43 |
| 98 | 18–35, 45–49 | | | 36–44 |
| 99 | 19–35, 37–38, 40–41, 43–44, 46–49 | | | 36, 39, 42, 45 |
| 100 | 20–35, 38, 41, 44, 47–50 | | | 36–37, 39–40, 42–43, 45–46 |

## LS[3](5, $k$, $v$)

166 of 928 admissible parameter sets exist, 762 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 14 | 7 | | | 6 |
| 15 | | | | 7 |
| 16 | 8 | | | |
| 23 | 6 | | | 7–8 |
| 24 | 7–8 | | | |
| 25 | 8 | | | |
| 32 | 9–10, 16 | | | 6–8, 11–15 |
| 33 | 7, 10 | | | 8–9, 11–16 |
| 34 | 17 | | | 8–16 |
| 35 | | | | 9–17 |
| 36 | | | | 10–18 |
| 37 | | | | 11–18 |
| 38 | | | | 12–19 |
| 39 | | | | 13–19 |
| 40 | 20 | | | 14–19 |
| 41 | 6 | | | 7–8, 15–20 |
| 42 | 16–17, 19–21 | | | 7–8, 18 |
| 43 | 17, 20–21 | | | 8, 18–19 |
| 44 | 21–22 | | | 18–20 |
| 45 | 19–20, 22 | | | 21 |
| 46 | 20, 23 | | | 21–22 |
| 47 | | | | 21–23 |
| 48 | 22–24 | | | |
| 49 | 23–24 | | | |
| 50 | 24–25 | | | 6–8, 15–17 |
| 51 | 25 | | | 7–8, 16–17 |
| 52 | 26 | | | 8, 17 |
| 59 | 6 | | | 7–26 |
| 60 | | | | 7–26 |
| 61 | | | | 8–26 |
| 62 | | | | 9–26 |
| 63 | | | | 10–26 |
| 64 | | | | 11–26 |
| 65 | | | | 12–26 |
| 66 | | | | 13–26 |
| 67 | | | | 14–26 |
| 68 | 34 | | | 6–8, 15–26, 33 |
| 69 | 16–20, 22–23, 25–26 | | | 7–8, 21, 24, 34 |
| 70 | 17–20, 23, 26, 35 | | | 8, 21–22, 24–25 |
| 71 | 18–20 | | | 21–26 |
| 72 | 19–26 | | | |
| 73 | 20–26 | | | |
| 74 | 21–26 | | | |
| 75 | 22–26 | | | |
| 76 | 23–26 | | | |
| 77 | 6, 24–26 | | | 7–8, 15–17, 33–35 |
| 78 | 25–26 | | | 7–8, 16–17, 34–35 |
| 79 | 26 | | | 8, 17, 35 |
| 86 | 43 | | | 6–42 |
| 87 | | | | 7–43 |
| 88 | 44 | | | 8–43 |
| 89 | | | | 9–44 |
| 90 | | | | 10–45 |
| 91 | | | | 11–45 |
| 92 | | | | 12–46 |
| 93 | | | | 13–46 |
| 94 | 47 | | | 14–46 |
| 95 | 6 | | | 7–8, 15–47 |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 96 | 16–17, 19–20, 22–23, 25–32, 43–44, 46–48 | | | 7–8, 18, 21, 24, 33–42, 45 |
| 97 | 17, 20, 23, 26–32, 44, 47–48 | | | 8, 18–19, 21–22, 24–25, 33–43, 45–46 |
| 98 | 27–32, 48–49 | | | 18–26, 33–47 |
| 99 | 19–23, 25–26, 28–32, 46–47, 49 | | | 24, 27, 33–45, 48 |
| 100 | 20–23, 26, 29–32, 47, 50 | | | 24–25, 27–28, 33–46, 48–49 |

## LS[3](6, $k$, $v$)

28 of 827 admissible parameter sets exist, 799 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 15 | | | | 7 |
| 16 | | | | 8 |
| 24 | | | | 7–8 |
| 25 | | | | 8 |
| 33 | 10 | | | 7–9, 11–16 |
| 34 | | | | 8–17 |
| 35 | | | | 9–17 |
| 36 | | | | 10–18 |
| 37 | | | | 11–18 |
| 38 | | | | 12–19 |
| 39 | | | | 13–19 |
| 40 | | | | 14–20 |
| 41 | | | | 15–20 |
| 42 | | | | 7–8, 16–21 |
| 43 | | | | 8, 17–21 |
| 44 | | | | 18–22 |
| 45 | 22 | | | 19–21 |
| 46 | 23 | | | 20–22 |
| 47 | | | | 21–23 |
| 48 | | | | 22–24 |
| 49 | | | | 23–24 |
| 50 | | | | 24–25 |
| 51 | 25 | | | 7–8, 16–17 |
| 52 | 26 | | | 8, 17 |
| 60 | | | | 7–26 |
| 61 | | | | 8–26 |
| 62 | | | | 9–26 |
| 63 | | | | 10–26 |
| 64 | | | | 11–26 |
| 65 | | | | 12–26 |
| 66 | | | | 13–26 |
| 67 | | | | 14–26 |
| 68 | | | | 15–26 |
| 69 | | | | 7–8, 16–26, 34 |
| 70 | | | | 8, 17–26, 35 |
| 71 | | | | 18–26 |
| 72 | 19–20 | | | 21–26 |
| 73 | 20 | | | 21–26 |
| 74 | | | | 21–26 |
| 75 | 22–23, 25–26 | | | 24 |
| 76 | 23, 26 | | | 24–25 |
| 77 | | | | 24–26 |
| 78 | 25–26 | | | 7–8, 16–17, 34–35 |
| 79 | 26 | | | 8, 17, 35 |
| 87 | | | | 7–43 |
| 88 | | | | 8–44 |
| 89 | | | | 9–44 |
| 90 | | | | 10–45 |
| 91 | | | | 11–45 |
| 92 | | | | 12–46 |
| 93 | | | | 13–46 |
| 94 | | | | 14–47 |
| 95 | | | | 15–47 |
| 96 | | | | 7–8, 16–48 |
| 97 | | | | 8, 17–48 |
| 98 | | | | 18–49 |
| 99 | 28–32, 49 | | | 19–27, 33–48 |
| 100 | 29–32, 50 | | | 20–28, 33–49 |

## LS[5](2, $k$, $v$)

631 of 922 admissible parameter sets exist, 1 do not exist, 290 are open.

| $v$ | exists, $k$ = | new | none | open, $k$ = |
|---|---|---|---|---|
| 7 | | | 3 | |
| 8 | 4 | | | |
| 12 | 3–4 | | | |
| 13 | 4 | | | |
| 17 | 3–4, 8 | | | |
| 18 | 4, 9 | | | |
| 22 | 3–4, 8–9 | | | |
| 23 | 4, 9 | | | |
| 27 | 3–8, 10–12 | | | 9, 13 |
| 28 | 4–8, 11–12 | | | 9–10, 13–14 |
| 29 | 5–8, 10–13 | | | 9, 14 |
| 30 | 6–8, 11–13 | | | 9–10, 14–15 |
| 31 | 7–14 | | | 15 |
| 32 | 3–4, 8–15 | | | 16 |
| 33 | 4, 9–15 | | | 16 |
| 34 | 10–15 | | | 16–17 |
| 35 | 11–17 | | | |
| 36 | 12–18 | | | |
| 37 | 3–4, 8, 13–18 | | | 9 |
| 38 | 4, 14–19 | | | 9 |
| 39 | 15–19 | | | |
| 40 | 16–20 | | | |
| 41 | 17–20 | | | |
| 42 | 3–4, 8, 18–21 | | | 9, 13–14 |
| 43 | 4, 19–21 | | | 9, 14 |
| 44 | 20–22 | | | |
| 45 | 21–22 | | | |
| 46 | 22–23 | | | |
| 47 | 3–4, 8–9, 13–14, 18–19, 23 | | | |
| 48 | 4, 9, 14, 19, 24 | | | |
| 52 | 3–8 | | | 9–24 |
| 53 | 4–11, 15–24 | | | 12–14 |
| 54 | 5–11, 16–24 | | | 12–15 |
| 55 | 6–11, 17–24 | | | 12–16 |
| 56 | 7–11, 18–24 | | | 12–17 |
| 57 | 3–4, 8–11, 19–24 | | | 12–18, 28 |
| 58 | 4, 9–11, 20–24 | | | 12–19, 29 |
| 59 | 10–24 | | | |
| 60 | 11–24 | | | |
| 61 | 12–24 | | | |
| 62 | 3–4, 8, 13–24 | | | 9, 28–29 |
| 63 | 4, 14–24 | | | 9, 29 |
| 64 | 15–24 | | | |
| 65 | 16–24 | | | |
| 66 | 17–24 | | | |
| 67 | 3–4, 8, 14, 18–24, 29 | | | 9, 13, 28, 33 |
| 68 | 4, 19–24 | | | 9, 14, 29, 34 |
| 69 | 20–24 | | | |
| 70 | 21–24 | | | |
| 71 | 22–24 | | | |
| 72 | 3–4, 8, 23–24 | | | 9, 13–14, 18–19, 28–29, 33–34 |
| 73 | 4, 14, 24 | | | 9, 19, 29, 34 |
| 77 | 3–8 | | | 9–24, 28–38 |
| 78 | 4–8 | | | 9–24, 29–39 |
| 79 | 5–8, 11, 17, 20, 23, 32, 35, 38 | | | 9–10, 12–16, 18–19, 21–22, 24, 30–31, 33–34, 36–37, 39 |
| 80 | 6–8 | | | 9–24, 31–40 |
| 81 | 7–8, 19, 22–23, 34, 38 | | | 9–18, 20–21, 24, 32–33, 35–37, 39–40 |
| 82 | 3–4, 8, 23 | | | 9–22, 24, 28–29, 33–41 |
| 83 | 4, 9–24, 29, 34–40 | | | 41 |
| 84 | 10–24, 35–40 | | | 41–42 |
| 85 | 11–24, 36–42 | | | |
| 86 | 12–24, 37–43 | | | |
| 87 | 3–4, 8, 13–24, 38–43 | | | 9, 28–29, 33–34 |
| 88 | 4, 14–24, 39–44 | | | 9, 29, 34 |
| 89 | 15–24, 40–44 | | | |
| 90 | 16–24, 41–45 | | | |
| 91 | 17–24, 42–45 | | | |
| 92 | 3–4, 8, 18–24, 43–46 | | | 9, 13–14, 28–29, 33–34, 38–39 |
| 93 | 4, 19–24, 44–46 | | | 9, 14, 29, 34, 39 |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 94 | 20–24, 45–47 | | | |
| 95 | 21–24, 46–47 | | | |
| 96 | 22–24, 47–48 | | | |
| 97 | 3–4, 8, 23–24, 38, 48 | | | 9, 13–14, 18–19, 28–29, 33–34, 39, 43–44 |
| 98 | 4, 24, 49 | | | 9, 14, 19, 29, 34, 39, 44 |

## LS[5]$(3, k, v)$

304 of 756 admissible parameter sets exist, 1 do not exist, 451 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 8 | | | 4 | |
| 13 | 4 | | | |
| 18 | 4, 9 | | | |
| 23 | 4, 9 | | | |
| 28 | 4–6, 8, 11–12 | | | 7, 9–10, 13–14 |
| 29 | 5–6, 12 | | | 7–11, 13–14 |
| 30 | 6, 11–13 | | | 7–10, 14–15 |
| 31 | 12–13 | | | 7–11, 14–15 |
| 32 | 9, 11, 13–14 | | | 8, 10, 12, 15–16 |
| 33 | 4, 9, 14 | | | 10–13, 15–16 |
| 34 | | | | 10–17 |
| 35 | | | | 11–17 |
| 36 | 18 | | | 12–17 |
| 37 | | | | 13–18 |
| 38 | 4, 14–19 | | | 9 |
| 39 | 15–19 | | | |
| 40 | 16–20 | | | |
| 41 | 17–20 | | | |
| 42 | 18–21 | | | |
| 43 | 4, 19–21 | | | 9, 14 |
| 44 | 20–22 | | | |
| 45 | 21–22 | | | |
| 46 | 22–23 | | | |
| 47 | 23 | | | |
| 48 | 4, 9, 14, 19, 24 | | | |
| 53 | 4–6 | | | 7–24 |
| 54 | 5–8, 10–11, 16–17, 19–24 | | | 9, 12–15, 18 |
| 55 | 6–8, 11, 17, 20–24 | | | 9–10, 12–16, 18–19 |
| 56 | 7–8, 21–24 | | | 9–20 |
| 57 | 8, 22–24 | | | 9–21 |
| 58 | 4, 23–24 | | | 9–22, 29 |
| 59 | 24 | | | 10–23 |
| 60 | 11–23 | | | 24 |
| 61 | 12–23 | | | 24 |
| 62 | 13–23 | | | 24 |
| 63 | 4, 14–23 | | | 9, 24, 29 |
| 64 | 15–23 | | | 24 |
| 65 | 16–24 | | | |
| 66 | 17–24 | | | |
| 67 | 18–24 | | | |
| 68 | 4, 19–24 | | | 9, 14, 29, 34 |
| 69 | 20–24 | | | |
| 70 | 21–24 | | | |
| 71 | 22–24 | | | |
| 72 | 23–24 | | | |
| 73 | 4, 24 | | | 9, 14, 19, 29, 34 |
| 78 | 4–6 | | | 7–24, 29–39 |
| 79 | 5–6 | | | 7–24, 30–39 |
| 80 | 6 | | | 7–24, 31–40 |
| 81 | | | | 7–24, 32–40 |
| 82 | | | | 8–24, 33–41 |
| 83 | 4 | | | 9–24, 29, 34–41 |
| 84 | 11, 13, 15, 17, 19, 23, 37, 39 | | | 10, 12, 14, 16, 18, 20–22, 24, 35–36, 38, 40–42 |
| 85 | | | | 11–24, 36–42 |
| 86 | 43 | | | 12–24, 37–42 |
| 87 | | | | 13–24, 38–43 |
| 88 | 4, 44 | | | 9, 14–24, 29, 34, 39–43 |
| 89 | | | | 15–24, 40–44 |
| 90 | 16–24, 41–45 | | | |
| 91 | 17–24, 42–45 | | | |
| 92 | 18–24, 43–46 | | | |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 93 | 4, 19–24, 44–46 | | | 9, 14, 29, 34, 39 |
| 94 | 20–24, 45–47 | | | |
| 95 | 21–24, 46–47 | | | |
| 96 | 22–24, 47–48 | | | |
| 97 | 23–24, 48 | | | |
| 98 | 4, 24, 49 | | | 9, 14, 19, 29, 34, 39, 44 |

## LS[5](4, $k$, $v$)

26 of 640 admissible parameter sets exist, 614 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 29 | | | | 5–14 |
| 30 | | | | 6–15 |
| 31 | | | | 7–15 |
| 32 | | | | 8–16 |
| 33 | | | | 9–16 |
| 34 | | | | 10–17 |
| 35 | | | | 11–17 |
| 36 | | | | 12–18 |
| 37 | | | | 13–18 |
| 38 | | | | 14–19 |
| 39 | | | | 15–19 |
| 40 | | | | 16–20 |
| 41 | | | | 17–20 |
| 42 | | | | 18–21 |
| 43 | | | | 19–21 |
| 44 | | | | 20–22 |
| 45 | | | | 21–22 |
| 46 | | | | 22–23 |
| 47 | | | | 23 |
| 48 | | | | 24 |
| 54 | | | | 5–24 |
| 55 | | | | 6–24 |
| 56 | | | | 7–24 |
| 57 | | | | 8–24 |
| 58 | | | | 9–24 |
| 59 | | | | 10–24 |
| 60 | | | | 11–24 |
| 61 | | | | 12–24 |
| 62 | | | | 13–24 |
| 63 | | | | 14–24 |
| 64 | | | | 15–24 |
| 65 | | | | 16–24 |
| 66 | | | | 17–24 |
| 67 | | | | 18–24 |
| 68 | | | | 19–24 |
| 69 | | | | 20–24 |
| 70 | 21–24 | | | |
| 71 | 22–24 | | | |
| 72 | 23–24 | | | |
| 73 | 24 | | | |
| 79 | | | | 5–24, 30–39 |
| 80 | | | | 6–24, 31–40 |
| 81 | | | | 7–24, 32–40 |
| 82 | | | | 8–24, 33–41 |
| 83 | | | | 9–24, 34–41 |
| 84 | | | | 10–24, 35–42 |
| 85 | | | | 11–24, 36–42 |
| 86 | | | | 12–24, 37–43 |
| 87 | | | | 13–24, 38–43 |
| 88 | | | | 14–24, 39–44 |
| 89 | | | | 15–24, 40–44 |
| 90 | | | | 16–24, 41–45 |
| 91 | | | | 17–24, 42–45 |
| 92 | | | | 18–24, 43–46 |
| 93 | | | | 19–24, 44–46 |
| 94 | | | | 20–24, 45–47 |
| 95 | 21–24, 46–47 | | | |
| 96 | 22–24, 47–48 | | | |
| 97 | 23–24, 48 | | | |
| 98 | 24, 49 | | | |

## LS[5](5, $k$, $v$)

2 of 580 admissible parameter sets exist, 578 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 30 | | | | 6–15 |
| 31 | | | | 7–15 |
| 32 | | | | 8–16 |
| 33 | | | | 9–16 |
| 34 | | | | 10–17 |
| 35 | | | | 11–17 |
| 36 | | | | 12–18 |
| 37 | | | | 13–18 |
| 38 | | | | 14–19 |
| 39 | | | | 15–19 |
| 40 | | | | 16–20 |
| 41 | | | | 17–20 |
| 42 | | | | 18–21 |
| 43 | | | | 19–21 |
| 44 | | | | 20–22 |
| 45 | | | | 21–22 |
| 46 | | | | 22–23 |
| 47 | | | | 23 |
| 48 | | | | 24 |
| 55 | | | | 6–24 |
| 56 | | | | 7–24 |
| 57 | | | | 8–24 |
| 58 | | | | 9–24 |
| 59 | | | | 10–24 |
| 60 | | | | 11–24 |
| 61 | | | | 12–24 |
| 62 | | | | 13–24 |
| 63 | | | | 14–24 |
| 64 | | | | 15–24 |
| 65 | | | | 16–24 |
| 66 | | | | 17–24 |
| 67 | | | | 18–24 |
| 68 | | | | 19–24 |
| 69 | | | | 20–24 |
| 70 | | | | 21–24 |
| 71 | | | | 22–24 |
| 72 | | | | 23–24 |
| 73 | | | | 24 |
| 80 | | | | 6–24, 31–40 |
| 81 | | | | 7–24, 32–40 |
| 82 | | | | 8–24, 33–41 |
| 83 | | | | 9–24, 34–41 |
| 84 | | | | 10–24, 35–42 |
| 85 | | | | 11–24, 36–42 |
| 86 | | | | 12–24, 37–43 |
| 87 | | | | 13–24, 38–43 |
| 88 | | | | 14–24, 39–44 |
| 89 | | | | 15–24, 40–44 |
| 90 | | | | 16–24, 41–45 |
| 91 | | | | 17–24, 42–45 |
| 92 | | | | 18–24, 43–46 |
| 93 | | | | 19–24, 44–46 |
| 94 | | | | 20–24, 45–47 |
| 95 | | | | 21–24, 46–47 |
| 96 | 48 | | | 22–24, 47 |
| 97 | | | | 23–24, 48 |
| 98 | 49 | | | 24 |

## LS[5](6, $k$, $v$)

None of the 522 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[7](2, $k$, $v$)

859 of 916 admissible parameter sets exist, 57 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 9 | 3–4 | | | |
| 10 | 4–5 | | | |
| 11 | 5 | | | |
| 12 | 6 | | | |
| 16 | 3–6 | | | |
| 17 | 4–6 | | | |
| 18 | 5–6 | | | |
| 19 | 6 | | | |
| 23 | 3–6, 10–11 | | | |
| 24 | 4–6, 11–12 | | | |
| 25 | 5–6, 12 | | | |
| 26 | 6, 13 | | | |
| 30 | 3–6, 10–13 | | | |
| 31 | 4–6, 11–13 | | | |
| 32 | 5–6, 12–13 | | | |
| 33 | 6, 13 | | | |
| 37 | 3–6, 10–13, 17–18 | | | |
| 38 | 4–6, 11–13, 18–19 | | | |
| 39 | 5–6, 12–13, 19 | | | |
| 40 | 6, 13, 20 | | | |
| 44 | 3–6, 10–13, 17–20 | | | |
| 45 | 4–6, 11–13, 18–20 | | | |
| 46 | 5–6, 12–13, 19–20 | | | |
| 47 | 6, 13, 20 | | | |
| 51 | 3–6, 10–13, 17–20, 24–25 | | | 7–9, 14–16, 21–23 |
| 52 | 4–6, 11–13, 18–20, 25–26 | | | 7–10, 14–17, 21–24 |
| 53 | 5–13, 15–26 | | | 14 |
| 54 | 6–13, 16–27 | | | 14–15 |
| 55 | 7–13, 17–27 | | | 14–16 |
| 56 | 8–13, 18–28 | | | 14–17 |
| 57 | 9–13, 19–28 | | | 14–18 |
| 58 | 3–6, 10–13, 17–29 | | | 14–16 |
| 59 | 4–6, 11–29 | | | |
| 60 | 5–6, 12–30 | | | |
| 61 | 6, 13–30 | | | |
| 62 | 14–31 | | | |
| 63 | 15–31 | | | |
| 64 | 16–32 | | | |
| 65 | 3–6, 10–13, 17–32 | | | |
| 66 | 4–6, 11–13, 18–33 | | | |
| 67 | 5–6, 12–13, 19–33 | | | |
| 68 | 6, 13, 20–34 | | | |
| 69 | 21–34 | | | |
| 70 | 22–35 | | | |
| 71 | 23–35 | | | |
| 72 | 3–6, 10–13, 17–20, 24–36 | | | |
| 73 | 4–6, 11–13, 18–20, 25–36 | | | |
| 74 | 5–6, 12–13, 19–20, 26–37 | | | |
| 75 | 6, 13, 20, 27–37 | | | |
| 76 | 28–38 | | | |
| 77 | 29–38 | | | |
| 78 | 30–39 | | | |
| 79 | 3–6, 10–13, 17–20, 24–27, 31–39 | | | |
| 80 | 4–6, 11–13, 18–20, 25–27, 32–40 | | | |
| 81 | 5–6, 12–13, 19–20, 26–27, 33–40 | | | |
| 82 | 6, 13, 20, 27, 34–41 | | | |
| 83 | 35–41 | | | |
| 84 | 36–42 | | | |
| 85 | 37–42 | | | |
| 86 | 3–6, 10–13, 17–20, 24–27, 31–34, 38–43 | | | |
| 87 | 4–6, 11–13, 18–20, 25–27, 32–34, 39–43 | | | |
| 88 | 5–6, 12–13, 19–20, 26–27, 33–34, 40–44 | | | |
| 89 | 6, 13, 20, 27, 34, 41–44 | | | |
| 90 | 42–45 | | | |
| 91 | 43–45 | | | |
| 92 | 44–46 | | | |
| 93 | 3–6, 10–13, 17–20, 24–27, 31–34, 38–41, 45–46 | | | |
| 94 | 4–6, 11–13, 18–20, 25–27, 32–34, 39–41, 46–47 | | | |
| 95 | 5–6, 12–13, 19–20, 26–27, 33–34, 40–41, 47 | | | |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 96 | 6, 13, 20, 27, 34, 41, 48 | | | |
| 100 | 3–6, 10–13, 17–20, 24–27, 31–34, 38–41, 45–48 | | | 7–9, 14–16, 21–23, 28–30, 35–37, 42–44 |

## LS[7]$(3, k, v)$

350 of 721 admissible parameter sets exist, 1 do not exist, 370 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 10 | 5 | | 4 | |
| 11 | 5 | | | |
| 12 | 6 | | | |
| 17 | 4–6 | | | |
| 18 | 5–6 | | | |
| 19 | 6 | | | |
| 24 | 4–6, 12 | | | 11 |
| 25 | 5–6 | | | 12 |
| 26 | 6, 13 | | | |
| 31 | 4–6 | | | 11–13 |
| 32 | 5–6, 12–13 | | | |
| 33 | 6, 13 | | | |
| 38 | 4–6, 12–13, 18–19 | | | 11 |
| 39 | 5–6, 13, 19 | | | 12 |
| 40 | 6, 20 | | | 13 |
| 45 | 4–6 | | | 11–13, 18–20 |
| 46 | 5–6 | | | 12–13, 19–20 |
| 47 | 6 | | | 13, 20 |
| 52 | 4–6, 26 | | | 7–25 |
| 53 | 5–6 | | | 7–26 |
| 54 | 6–8, 10–11, 16–17, 19–25, 27 | | | 9, 12–15, 18, 26 |
| 55 | 7–8, 11, 17, 20–25 | | | 9–10, 12–16, 18–19, 26–27 |
| 56 | 8, 21–25, 28 | | | 9–20, 26–27 |
| 57 | 22–25 | | | 9–21, 26–28 |
| 58 | 23–25, 29 | | | 10–22, 26–28 |
| 59 | 4–6, 24–25 | | | 11–23, 26–29 |
| 60 | 5–6, 12–17, 19–23, 25–28, 30 | | | 18, 24, 29 |
| 61 | 6, 13–17, 20–23, 26–28 | | | 18–19, 24–25, 29–30 |
| 62 | 14–17, 21–23, 27–28, 31 | | | 18–20, 24–26, 29–30 |
| 63 | 15–17, 22–23, 28 | | | 18–21, 24–27, 29–31 |
| 64 | 16–17, 23, 32 | | | 18–22, 24–31 |
| 65 | 17 | | | 18–32 |
| 66 | 4–6, 33 | | | 11–13, 18–32 |
| 67 | 5–6 | | | 12–13, 19–33 |
| 68 | 6, 21, 25, 34 | | | 13, 20, 22–24, 26–33 |
| 69 | | | | 21–34 |
| 70 | 22–27, 29–35 | | | 28 |
| 71 | 23–27, 30–35 | | | 28–29 |
| 72 | 24–27, 31–36 | | | 28–30 |
| 73 | 4–6, 25–27, 32–36 | | | 11–13, 18–20, 28–31 |
| 74 | 5–6, 26–27, 33–37 | | | 12–13, 19–20, 28–32 |
| 75 | 6, 27, 34–37 | | | 13, 20, 28–33 |
| 76 | 35–38 | | | 28–34 |
| 77 | 29–38 | | | |
| 78 | 30–39 | | | |
| 79 | 31–39 | | | |
| 80 | 4–6, 32–40 | | | 11–13, 18–20, 25–27 |
| 81 | 5–6, 33–40 | | | 12–13, 19–20, 26–27 |
| 82 | 6, 34–41 | | | 13, 20, 27 |
| 83 | 35–41 | | | |
| 84 | 36–42 | | | |
| 85 | 37–42 | | | |
| 86 | 38–43 | | | |
| 87 | 4–6, 39–43 | | | 11–13, 18–20, 25–27, 32–34 |
| 88 | 5–6, 40–44 | | | 12–13, 19–20, 26–27, 33–34 |
| 89 | 6, 41–44 | | | 13, 20, 27, 34 |
| 90 | 42–45 | | | |
| 91 | 43–45 | | | |
| 92 | 44–46 | | | |
| 93 | 45–46 | | | |
| 94 | 4–6, 46–47 | | | 11–13, 18–20, 25–27, 32–34, 39–41 |
| 95 | 5–6, 47 | | | 12–13, 19–20, 26–27, 33–34, 40–41 |
| 96 | 6, 48 | | | 13, 20, 27, 34, 41 |

## LS[7](4, $k$, $v$)

48 of 602 admissible parameter sets exist, 1 do not exist, 553 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 11 | | | 5 | |
| 12 | | | | 6 |
| 18 | | | | 5–6 |
| 19 | | | | 6 |
| 25 | | | | 5–6, 12 |
| 26 | | | | 6, 13 |
| 32 | 5 | | | 6, 12–13 |
| 33 | | | | 6, 13 |
| 39 | | | | 5–6, 12–13, 19 |
| 40 | | | | 6, 13, 20 |
| 46 | | | | 5–6, 12–13, 19–20 |
| 47 | | | | 6, 13, 20 |
| 53 | | | | 5–26 |
| 54 | | | | 6–27 |
| 55 | | | | 7–27 |
| 56 | | | | 8–28 |
| 57 | | | | 9–28 |
| 58 | | | | 10–29 |
| 59 | | | | 11–29 |
| 60 | 5 | | | 6, 12–30 |
| 61 | | | | 6, 13–30 |
| 62 | | | | 14–31 |
| 63 | | | | 15–31 |
| 64 | | | | 16–32 |
| 65 | | | | 17–32 |
| 66 | | | | 18–33 |
| 67 | | | | 5–6, 12–13, 19–33 |
| 68 | | | | 6, 13, 20–34 |
| 69 | | | | 21–34 |
| 70 | | | | 22–35 |
| 71 | | | | 23–35 |
| 72 | | | | 24–36 |
| 73 | | | | 25–36 |
| 74 | | | | 5–6, 12–13, 19–20, 26–37 |
| 75 | | | | 6, 13, 20, 27–37 |
| 76 | | | | 28–38 |
| 77 | 36–38 | | | 29–35 |
| 78 | 37–39 | | | 30–36 |
| 79 | 38–39 | | | 31–37 |
| 80 | 39–40 | | | 32–38 |
| 81 | 40 | | | 5–6, 12–13, 19–20, 26–27, 33–39 |
| 82 | 41 | | | 6, 13, 20, 27, 34–40 |
| 83 | | | | 35–41 |
| 84 | 36–41 | | | 42 |
| 85 | 37–41 | | | 42 |
| 86 | 38–41 | | | 42–43 |
| 87 | 39–41 | | | 42–43 |
| 88 | 5, 40–41 | | | 6, 12–13, 19–20, 26–27, 33–34, 42–44 |
| 89 | 41 | | | 6, 13, 20, 27, 34, 42–44 |
| 90 | | | | 42–45 |
| 91 | 43–45 | | | |
| 92 | 44–46 | | | |
| 93 | 45–46 | | | |
| 94 | 46–47 | | | |
| 95 | 47 | | | 5–6, 12–13, 19–20, 26–27, 33–34, 40–41 |
| 96 | 48 | | | 6, 13, 20, 27, 34, 41 |

## LS[7](5, $k$, $v$)

6 of 517 admissible parameter sets exist, 1 do not exist, 510 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 12 | | | 6 | |
| 19 | | | | 6 |
| 26 | | | | 6, 13 |
| 33 | | | | 6, 13 |
| 40 | | | | 6, 13, 20 |
| 47 | | | | 6, 13, 20 |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 54 | | | | 6–27 |
| 55 | | | | 7–27 |
| 56 | | | | 8–28 |
| 57 | | | | 9–28 |
| 58 | | | | 10–29 |
| 59 | | | | 11–29 |
| 60 | | | | 12–30 |
| 61 | | | | 6, 13–30 |
| 62 | | | | 14–31 |
| 63 | | | | 15–31 |
| 64 | | | | 16–32 |
| 65 | | | | 17–32 |
| 66 | | | | 18–33 |
| 67 | | | | 19–33 |
| 68 | | | | 6, 13, 20–34 |
| 69 | | | | 21–34 |
| 70 | | | | 22–35 |
| 71 | | | | 23–35 |
| 72 | | | | 24–36 |
| 73 | | | | 25–36 |
| 74 | | | | 26–37 |
| 75 | | | | 6, 13, 20, 27–37 |
| 76 | | | | 28–38 |
| 77 | | | | 29–38 |
| 78 | 39 | | | 30–38 |
| 79 | | | | 31–39 |
| 80 | 40 | | | 32–39 |
| 81 | | | | 33–40 |
| 82 | 41 | | | 6, 13, 20, 27, 34–40 |
| 83 | | | | 35–41 |
| 84 | | | | 36–42 |
| 85 | | | | 37–42 |
| 86 | | | | 38–43 |
| 87 | | | | 39–43 |
| 88 | | | | 40–44 |
| 89 | | | | 6, 13, 20, 27, 34, 41–44 |
| 90 | | | | 42–45 |
| 91 | | | | 43–45 |
| 92 | 46 | | | 44–45 |
| 93 | | | | 45–46 |
| 94 | 47 | | | 46 |
| 95 | | | | 47 |
| 96 | 48 | | | 6, 13, 20, 27, 34, 41 |

## LS[7](6, $k$, $v$)

None of the 462 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[11](2, $k$, $v$)

All 656 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ are realised.

## LS[11](3, $k$, $v$)

69 of 512 admissible parameter sets exist, 443 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 14 | 7 | | | 4–6 |
| 15 | | | | 5–7 |
| 16 | 8 | | | 6–7 |
| 17 | | | | 7–8 |
| 18 | 9 | | | 8 |
| 19 | | | | 9 |
| 20 | 10 | | | |
| 25 | | | | 4–10 |
| 26 | | | | 5–10 |
| 27 | | | | 6–10 |
| 28 | | | | 7–10 |
| 29 | | | | 8–10 |
| 30 | | | | 9–10 |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 31 | | | | 10 |
| 36 | 4, 18 | | | 5–10, 15–17 |
| 37 | | | | 5–10, 16–18 |
| 38 | 17–19 | | | 6–10 |
| 39 | 18–19 | | | 7–10 |
| 40 | 19–20 | | | 8–10 |
| 41 | 20 | | | 9–10 |
| 42 | 21 | | | 10 |
| 47 | | **4** | | 5–10, 15–21 |
| 48 | 5–7, 9–10, 17–21 | | | 8, 16 |
| 49 | 6–7, 10, 18–21 | | | 8–9, 17 |
| 50 | 7, 19–21 | | | 8–10, 18 |
| 51 | 20–21 | | | 8–10, 19 |
| 52 | 21 | | | 9–10, 20 |
| 53 | | | | 10, 21 |
| 58 | 29 | | | 4–10, 15–21, 26–28 |
| 59 | | | | 5–10, 16–21, 27–29 |
| 60 | 7–9, 17, 19–21, 30 | | | 6, 10, 18, 28–29 |
| 61 | 8–9, 20–21 | | | 7, 10, 18–19, 29–30 |
| 62 | 9, 21, 31 | | | 8, 10, 19–20, 30 |
| 63 | | | | 9–10, 20–21, 31 |
| 64 | 32 | | | 10, 21 |
| 69 | 4 | | | 5–10, 15–21, 26–32 |
| 70 | | | | 5–10, 16–21, 27–32 |
| 71 | | | | 6–10, 17–21, 28–32 |
| 72 | 19, 29 | | | 7–10, 18, 20–21, 30–32 |
| 73 | | | | 8–10, 19–21, 30–32 |
| 74 | | | | 9–10, 20–21, 31–32 |
| 75 | | | | 10, 21, 32 |
| 80 | 40 | **4** | | 5–10, 15–21, 26–32, 37–39 |
| 81 | | | | 5–10, 16–21, 27–32, 38–40 |
| 82 | 41 | | | 6–10, 17–21, 28–32, 39–40 |
| 83 | | | | 7–10, 18–21, 29–32, 40–41 |
| 84 | 9, 19, 31, 42 | | | 8, 10, 20–21, 30, 32, 41 |
| 85 | | | | 9–10, 20–21, 31–32, 42 |
| 86 | 43 | | | 10, 21, 32 |
| 91 | | **4** | | 5–10, 15–21, 26–32, 37–43 |
| 92 | | | | 5–10, 16–21, 27–32, 38–43 |
| 93 | | | | 6–10, 17–21, 28–32, 39–43 |
| 94 | | | | 7–10, 18–21, 29–32, 40–43 |
| 95 | | | | 8–10, 19–21, 30–32, 41–43 |
| 96 | | | | 9–10, 20–21, 31–32, 42–43 |
| 97 | | | | 10, 21, 32, 43 |

## LS[11]$(4, k, v)$

0 of 384 admissible parameter sets exist, 1 do not exist, 383 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 15 | | | 5 | 6–7 |
| 16 | | | | 6–8 |
| 17 | | | | 7–8 |
| 18 | | | | 8–9 |
| 19 | | | | 9 |
| 20 | | | | 10 |
| 26 | | | | 5–10 |
| 27 | | | | 6–10 |
| 28 | | | | 7–10 |
| 29 | | | | 8–10 |
| 30 | | | | 9–10 |
| 31 | | | | 10 |
| 37 | | | | 5–10, 16–18 |
| 38 | | | | 6–10, 17–19 |
| 39 | | | | 7–10, 18–19 |
| 40 | | | | 8–10, 19–20 |
| 41 | | | | 9–10, 20 |
| 42 | | | | 10, 21 |
| 48 | | | | 5–10, 16–21 |
| 49 | | | | 6–10, 17–21 |
| 50 | | | | 7–10, 18–21 |
| 51 | | | | 8–10, 19–21 |
| 52 | | | | 9–10, 20–21 |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 53 | | | | 10, 21 |
| 59 | | | | 5–10, 16–21, 27–29 |
| 60 | | | | 6–10, 17–21, 28–30 |
| 61 | | | | 7–10, 18–21, 29–30 |
| 62 | | | | 8–10, 19–21, 30–31 |
| 63 | | | | 9–10, 20–21, 31 |
| 64 | | | | 10, 21, 32 |
| 70 | | | | 5–10, 16–21, 27–32 |
| 71 | | | | 6–10, 17–21, 28–32 |
| 72 | | | | 7–10, 18–21, 29–32 |
| 73 | | | | 8–10, 19–21, 30–32 |
| 74 | | | | 9–10, 20–21, 31–32 |
| 75 | | | | 10, 21, 32 |
| 81 | | | | 5–10, 16–21, 27–32, 38–40 |
| 82 | | | | 6–10, 17–21, 28–32, 39–41 |
| 83 | | | | 7–10, 18–21, 29–32, 40–41 |
| 84 | | | | 8–10, 19–21, 30–32, 41–42 |
| 85 | | | | 9–10, 20–21, 31–32, 42 |
| 86 | | | | 10, 21, 32, 43 |
| 92 | | | | 5–10, 16–21, 27–32, 38–43 |
| 93 | | | | 6–10, 17–21, 28–32, 39–43 |
| 94 | | | | 7–10, 18–21, 29–32, 40–43 |
| 95 | | | | 8–10, 19–21, 30–32, 41–43 |
| 96 | | | | 9–10, 20–21, 31–32, 42–43 |
| 97 | | | | 10, 21, 32, 43 |

## LS[11]$(5, k, v)$

0 of 276 admissible parameter sets exist, 1 do not exist, 275 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 16 | | | 6 | 7–8 |
| 17 | | | | 7–8 |
| 18 | | | | 8–9 |
| 19 | | | | 9 |
| 20 | | | | 10 |
| 27 | | | | 6–10 |
| 28 | | | | 7–10 |
| 29 | | | | 8–10 |
| 30 | | | | 9–10 |
| 31 | | | | 10 |
| 38 | | | | 6–10, 17–19 |
| 39 | | | | 7–10, 18–19 |
| 40 | | | | 8–10, 19–20 |
| 41 | | | | 9–10, 20 |
| 42 | | | | 10, 21 |
| 49 | | | | 6–10, 17–21 |
| 50 | | | | 7–10, 18–21 |
| 51 | | | | 8–10, 19–21 |
| 52 | | | | 9–10, 20–21 |
| 53 | | | | 10, 21 |
| 60 | | | | 6–10, 17–21, 28–30 |
| 61 | | | | 7–10, 18–21, 29–30 |
| 62 | | | | 8–10, 19–21, 30–31 |
| 63 | | | | 9–10, 20–21, 31 |
| 64 | | | | 10, 21, 32 |
| 71 | | | | 6–10, 17–21, 28–32 |
| 72 | | | | 7–10, 18–21, 29–32 |
| 73 | | | | 8–10, 19–21, 30–32 |
| 74 | | | | 9–10, 20–21, 31–32 |
| 75 | | | | 10, 21, 32 |
| 82 | | | | 6–10, 17–21, 28–32, 39–41 |
| 83 | | | | 7–10, 18–21, 29–32, 40–41 |
| 84 | | | | 8–10, 19–21, 30–32, 41–42 |
| 85 | | | | 9–10, 20–21, 31–32, 42 |
| 86 | | | | 10, 21, 32, 43 |
| 93 | | | | 6–10, 17–21, 28–32, 39–43 |
| 94 | | | | 7–10, 18–21, 29–32, 40–43 |
| 95 | | | | 8–10, 19–21, 30–32, 41–43 |
| 96 | | | | 9–10, 20–21, 31–32, 42–43 |
| 97 | | | | 10, 21, 32, 43 |

## LS[11](6, $k$, $v$)

0 of 184 admissible parameter sets exist, 1 do not exist, 183 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 17 | | | 7 | 8 |
| 18 | | | | 8–9 |
| 19 | | | | 9 |
| 20 | | | | 10 |
| 28 | | | | 7–10 |
| 29 | | | | 8–10 |
| 30 | | | | 9–10 |
| 31 | | | | 10 |
| 39 | | | | 7–10, 18–19 |
| 40 | | | | 8–10, 19–20 |
| 41 | | | | 9–10, 20 |
| 42 | | | | 10, 21 |
| 50 | | | | 7–10, 18–21 |
| 51 | | | | 8–10, 19–21 |
| 52 | | | | 9–10, 20–21 |
| 53 | | | | 10, 21 |
| 61 | | | | 7–10, 18–21, 29–30 |
| 62 | | | | 8–10, 19–21, 30–31 |
| 63 | | | | 9–10, 20–21, 31 |
| 64 | | | | 10, 21, 32 |
| 72 | | | | 7–10, 18–21, 29–32 |
| 73 | | | | 8–10, 19–21, 30–32 |
| 74 | | | | 9–10, 20–21, 31–32 |
| 75 | | | | 10, 21, 32 |
| 83 | | | | 7–10, 18–21, 29–32, 40–41 |
| 84 | | | | 8–10, 19–21, 30–32, 41–42 |
| 85 | | | | 9–10, 20–21, 31–32, 42 |
| 86 | | | | 10, 21, 32, 43 |
| 94 | | | | 7–10, 18–21, 29–32, 40–43 |
| 95 | | | | 8–10, 19–21, 30–32, 41–43 |
| 96 | | | | 9–10, 20–21, 31–32, 42–43 |
| 97 | | | | 10, 21, 32, 43 |

## LS[13](2, $k$, $v$)

286 of 769 admissible parameter sets exist, 483 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 15 | 3–4, 7 | | | 5–6 |
| 16 | 4, 8 | | | 5–7 |
| 17 | | | | 5–8 |
| 18 | | | | 6–9 |
| 19 | 8 | | | 7, 9 |
| 20 | | | | 8–10 |
| 21 | | | | 9–10 |
| 22 | | | | 10–11 |
| 23 | | | | 11 |
| 24 | | | | 12 |
| 28 | 3–4 | | | 5–12 |
| 29 | 4, 6–11 | | | 5, 12 |
| 30 | 7–11 | | | 5–6, 12 |
| 31 | 7–12 | | | 6 |
| 32 | 7–12 | | | |
| 33 | 8–12 | | | |
| 34 | 9–12 | | | |
| 35 | 10–12 | | | |
| 36 | 11–12 | | | |
| 37 | 12 | | | |
| 41 | 3–4, 18–19 | | | 5–12, 16–17, 20 |
| 42 | 4, 19 | | | 5–12, 17–18, 20–21 |
| 43 | 5, 9–12, 18–20 | | | 6–8, 21 |
| 44 | 10–12, 19–20 | | | 6–9, 21–22 |
| 45 | 11–12, 20 | | | 7–10, 21–22 |
| 46 | 12 | | | 8–11, 21–23 |
| 47 | 9–12, 22 | | | 23 |
| 48 | 10–12 | | | 23–24 |
| 49 | 11–12 | | | 24 |
| 50 | 12 | | | 25 |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 54 | 3–4 | | | 5–12, 16–25 |
| 55 | 4 | | | 5–12, 17–25 |
| 56 | | | | 5–12, 18–25 |
| 57 | | | | 6–12, 19–25 |
| 58 | | | | 7–12, 20–25 |
| 59 | 8–12, 21–25 | | | |
| 60 | 9–12, 22–25 | | | |
| 61 | 10–12, 23–25 | | | |
| 62 | 11–12, 24–25 | | | |
| 63 | 12, 25 | | | |
| 67 | 3–8, 16–21, 24–25, 29, 32 | | | 9–12, 22–23, 30–31, 33 |
| 68 | 4–8, 17–21, 25 | | | 9–12, 22–24, 30–34 |
| 69 | 5–8, 18–21 | | | 9–12, 22–25, 31–34 |
| 70 | 6–8, 19–21 | | | 9–12, 22–25, 32–35 |
| 71 | 7–9, 12, 20–21, 23–24, 33–34 | | | 10–11, 22, 25, 35 |
| 72 | 8–9, 21, 24, 34 | | | 10–12, 22–23, 25, 35–36 |
| 73 | 9, 11, 23, 35 | | | 10, 12, 22, 24–25, 36 |
| 74 | | | | 10–12, 23–25, 36–37 |
| 75 | | | | 11–12, 24–25, 37 |
| 76 | | | | 12, 25, 38 |
| 80 | 3–4, 7–8 | | | 5–6, 9–12, 16–25, 29–38 |
| 81 | 4, 8, 19, 22–23, 34, 38 | | | 5–7, 9–12, 17–18, 20–21, 24–25, 30–33, 35–37 |
| 82 | 23 | | | 5–12, 18–22, 24–25, 31–38 |
| 83 | 6–12, 19–25, 32–38 | | | |
| 84 | 7–12, 20–25, 33–38 | | | |
| 85 | 8–12, 21–25, 34–38 | | | |
| 86 | 9–12, 22–25, 35–38 | | | |
| 87 | 10–12, 23–25, 36–38 | | | |
| 88 | 11–12, 24–25, 37–38 | | | |
| 89 | 12, 25, 38 | | | |
| 93 | 3–4 | | | 5–12, 16–25, 29–38, 42–46 |
| 94 | 4, 6–8 | | | 5, 9–12, 17–25, 30–38, 43–47 |
| 95 | 7–8 | | | 5–6, 9–12, 18–25, 31–38, 44–47 |
| 96 | 7–8 | | | 6, 9–12, 19–25, 32–38, 45–48 |
| 97 | 7–8, 35, 38, 47 | | | 9–12, 20–25, 33–34, 36–37, 46, 48 |
| 98 | 8 | | | 9–12, 21–25, 34–38, 47–49 |
| 99 | | | | 9–12, 22–25, 35–38, 48–49 |
| 100 | | | | 10–12, 23–25, 36–38, 49–50 |

## LS[13]$(3, k, v)$

52 of 629 admissible parameter sets exist, 577 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 16 | 8 | | | 4–7 |
| 17 | | | | 5–8 |
| 18 | | | | 6–9 |
| 19 | | | | 7–9 |
| 20 | | | | 8–10 |
| 21 | | | | 9–10 |
| 22 | | | | 10–11 |
| 23 | | | | 11 |
| 24 | | | | 12 |
| 29 | | | | 4–12 |
| 30 | 7–11 | | | 5–6, 12 |
| 31 | 8–11 | | | 6–7, 12 |
| 32 | 9–12 | | | 7–8 |
| 33 | 10–12 | | | 8–9 |
| 34 | 11–12 | | | 9–10 |
| 35 | 12 | | | 10–11 |
| 36 | | | | 11–12 |
| 37 | | | | 12 |
| 42 | 4, 19 | | | 5–12, 17–18, 20–21 |
| 43 | | | | 5–12, 18–21 |
| 44 | 10, 12, 19–20 | | | 6–9, 11, 21–22 |
| 45 | 20 | | | 7–12, 21–22 |
| 46 | | | | 8–12, 21–23 |
| 47 | | | | 9–12, 22–23 |
| 48 | 10–12 | | | 23–24 |
| 49 | 11–12 | | | 24 |
| 50 | 12 | | | 25 |
| 55 | | | | 4–12, 17–25 |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 56 | | | | 5–12, 18–25 |
| 57 | | | | 6–12, 19–25 |
| 58 | | | | 7–12, 20–25 |
| 59 | | | | 8–12, 21–25 |
| 60 | 9, 11–12, 22–23, 25 | | | 10, 24 |
| 61 | 12, 23 | | | 10–11, 24–25 |
| 62 | | | | 11–12, 24–25 |
| 63 | | | | 12, 25 |
| 68 | 7, 19, 21, 25 | | | 4–6, 8–12, 17–18, 20, 22–24, 30–34 |
| 69 | | | | 5–12, 18–25, 31–34 |
| 70 | | | | 6–12, 19–25, 32–35 |
| 71 | | | | 7–12, 20–25, 33–35 |
| 72 | | | | 8–12, 21–25, 34–36 |
| 73 | | | | 9–12, 22–25, 35–36 |
| 74 | | | | 10–12, 23–25, 36–37 |
| 75 | | | | 11–12, 24–25, 37 |
| 76 | | | | 12, 25, 38 |
| 81 | 4 | | | 5–12, 17–25, 30–38 |
| 82 | | | | 5–12, 18–25, 31–38 |
| 83 | | | | 6–12, 19–25, 32–38 |
| 84 | 9, 11, 23, 25, 33, 37 | | | 7–8, 10, 12, 20–22, 24, 34–36, 38 |
| 85 | | | | 8–12, 21–25, 34–38 |
| 86 | | | | 9–12, 22–25, 35–38 |
| 87 | | | | 10–12, 23–25, 36–38 |
| 88 | | | | 11–12, 24–25, 37–38 |
| 89 | | | | 12, 25, 38 |
| 94 | | | | 4–12, 17–25, 30–38, 43–47 |
| 95 | | | | 5–12, 18–25, 31–38, 44–47 |
| 96 | | | | 6–12, 19–25, 32–38, 45–48 |
| 97 | | | | 7–12, 20–25, 33–38, 46–48 |
| 98 | | | | 8–12, 21–25, 34–38, 47–49 |
| 99 | | | | 9–12, 22–25, 35–38, 48–49 |
| 100 | | | | 10–12, 23–25, 36–38, 49–50 |

## LS[13]$(4, k, v)$

6 of 501 admissible parameter sets exist, 1 do not exist, 494 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 17 | | | 5 | 6–8 |
| 18 | | | | 6–9 |
| 19 | | | | 7–9 |
| 20 | | | | 8–10 |
| 21 | | | | 9–10 |
| 22 | | | | 10–11 |
| 23 | | | | 11 |
| 24 | | | | 12 |
| 30 | | | | 5–12 |
| 31 | 8–10 | | | 6–7, 11–12 |
| 32 | 9–10 | | | 7–8, 11–12 |
| 33 | 10 | | | 8–9, 11–12 |
| 34 | | | | 9–12 |
| 35 | | | | 10–12 |
| 36 | | | | 11–12 |
| 37 | | | | 12 |
| 43 | | | | 5–12, 18–21 |
| 44 | | | | 6–12, 19–22 |
| 45 | | | | 7–12, 20–22 |
| 46 | | | | 8–12, 21–23 |
| 47 | | | | 9–12, 22–23 |
| 48 | | | | 10–12, 23–24 |
| 49 | | | | 11–12, 24 |
| 50 | | | | 12, 25 |
| 56 | | | | 5–12, 18–25 |
| 57 | | | | 6–12, 19–25 |
| 58 | | | | 7–12, 20–25 |
| 59 | | | | 8–12, 21–25 |
| 60 | | | | 9–12, 22–25 |
| 61 | | | | 10–12, 23–25 |
| 62 | | | | 11–12, 24–25 |
| 63 | | | | 12, 25 |
| 69 | | | | 5–12, 18–25, 31–34 |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 70 | | | | 6–12, 19–25, 32–35 |
| 71 | | | | 7–12, 20–25, 33–35 |
| 72 | | | | 8–12, 21–25, 34–36 |
| 73 | | | | 9–12, 22–25, 35–36 |
| 74 | | | | 10–12, 23–25, 36–37 |
| 75 | | | | 11–12, 24–25, 37 |
| 76 | | | | 12, 25, 38 |
| 82 | | | | 5–12, 18–25, 31–38 |
| 83 | | | | 6–12, 19–25, 32–38 |
| 84 | | | | 7–12, 20–25, 33–38 |
| 85 | | | | 8–12, 21–25, 34–38 |
| 86 | | | | 9–12, 22–25, 35–38 |
| 87 | | | | 10–12, 23–25, 36–38 |
| 88 | | | | 11–12, 24–25, 37–38 |
| 89 | | | | 12, 25, 38 |
| 95 | | | | 5–12, 18–25, 31–38, 44–47 |
| 96 | | | | 6–12, 19–25, 32–38, 45–48 |
| 97 | | | | 7–12, 20–25, 33–38, 46–48 |
| 98 | | | | 8–12, 21–25, 34–38, 47–49 |
| 99 | | | | 9–12, 22–25, 35–38, 48–49 |
| 100 | | | | 10–12, 23–25, 36–38, 49–50 |

## LS[13]$(5, k, v)$

3 of 389 admissible parameter sets exist, 1 do not exist, 385 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 18 | | | 6 | 7–9 |
| 19 | | | | 7–9 |
| 20 | | | | 8–10 |
| 21 | | | | 9–10 |
| 22 | | | | 10–11 |
| 23 | | | | 11 |
| 24 | | | | 12 |
| 31 | | | | 6–12 |
| 32 | 9–10 | | | 7–8, 11–12 |
| 33 | 10 | | | 8–9, 11–12 |
| 34 | | | | 9–12 |
| 35 | | | | 10–12 |
| 36 | | | | 11–12 |
| 37 | | | | 12 |
| 44 | | | | 6–12, 19–22 |
| 45 | | | | 7–12, 20–22 |
| 46 | | | | 8–12, 21–23 |
| 47 | | | | 9–12, 22–23 |
| 48 | | | | 10–12, 23–24 |
| 49 | | | | 11–12, 24 |
| 50 | | | | 12, 25 |
| 57 | | | | 6–12, 19–25 |
| 58 | | | | 7–12, 20–25 |
| 59 | | | | 8–12, 21–25 |
| 60 | | | | 9–12, 22–25 |
| 61 | | | | 10–12, 23–25 |
| 62 | | | | 11–12, 24–25 |
| 63 | | | | 12, 25 |
| 70 | | | | 6–12, 19–25, 32–35 |
| 71 | | | | 7–12, 20–25, 33–35 |
| 72 | | | | 8–12, 21–25, 34–36 |
| 73 | | | | 9–12, 22–25, 35–36 |
| 74 | | | | 10–12, 23–25, 36–37 |
| 75 | | | | 11–12, 24–25, 37 |
| 76 | | | | 12, 25, 38 |
| 83 | | | | 6–12, 19–25, 32–38 |
| 84 | | | | 7–12, 20–25, 33–38 |
| 85 | | | | 8–12, 21–25, 34–38 |
| 86 | | | | 9–12, 22–25, 35–38 |
| 87 | | | | 10–12, 23–25, 36–38 |
| 88 | | | | 11–12, 24–25, 37–38 |
| 89 | | | | 12, 25, 38 |
| 96 | | | | 6–12, 19–25, 32–38, 45–48 |
| 97 | | | | 7–12, 20–25, 33–38, 46–48 |
| 98 | | | | 8–12, 21–25, 34–38, 47–49 |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 99 | | | | 9–12, 22–25, 35–38, 48–49 |
| 100 | | | | 10–12, 23–25, 36–38, 49–50 |

## LS[13](6, $k$, $v$)

1 of 289 admissible parameter sets exist, 1 do not exist, 287 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 19 | | | 7 | 8–9 |
| 20 | | | | 8–10 |
| 21 | | | | 9–10 |
| 22 | | | | 10–11 |
| 23 | | | | 11 |
| 24 | | | | 12 |
| 32 | | | | 7–12 |
| 33 | 10 | | | 8–9, 11–12 |
| 34 | | | | 9–12 |
| 35 | | | | 10–12 |
| 36 | | | | 11–12 |
| 37 | | | | 12 |
| 45 | | | | 7–12, 20–22 |
| 46 | | | | 8–12, 21–23 |
| 47 | | | | 9–12, 22–23 |
| 48 | | | | 10–12, 23–24 |
| 49 | | | | 11–12, 24 |
| 50 | | | | 12, 25 |
| 58 | | | | 7–12, 20–25 |
| 59 | | | | 8–12, 21–25 |
| 60 | | | | 9–12, 22–25 |
| 61 | | | | 10–12, 23–25 |
| 62 | | | | 11–12, 24–25 |
| 63 | | | | 12, 25 |
| 71 | | | | 7–12, 20–25, 33–35 |
| 72 | | | | 8–12, 21–25, 34–36 |
| 73 | | | | 9–12, 22–25, 35–36 |
| 74 | | | | 10–12, 23–25, 36–37 |
| 75 | | | | 11–12, 24–25, 37 |
| 76 | | | | 12, 25, 38 |
| 84 | | | | 7–12, 20–25, 33–38 |
| 85 | | | | 8–12, 21–25, 34–38 |
| 86 | | | | 9–12, 22–25, 35–38 |
| 87 | | | | 10–12, 23–25, 36–38 |
| 88 | | | | 11–12, 24–25, 37–38 |
| 89 | | | | 12, 25, 38 |
| 97 | | | | 7–12, 20–25, 33–38, 46–48 |
| 98 | | | | 8–12, 21–25, 34–38, 47–49 |
| 99 | | | | 9–12, 22–25, 35–38, 48–49 |
| 100 | | | | 10–12, 23–25, 36–38, 49–50 |

## LS[17](2, $k$, $v$)

304 of 798 admissible parameter sets exist, 494 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 19 | 3–8 | | | 9 |
| 20 | 4–8 | | | 9–10 |
| 21 | 5–8 | | | 9–10 |
| 22 | 6–8 | | | 9–11 |
| 23 | 7–10 | | | 11 |
| 24 | 8–10 | | | 11–12 |
| 25 | 9–11 | | | 12 |
| 26 | 10–11 | | | 12–13 |
| 27 | 11 | | | 12–13 |
| 28 | | | | 12–14 |
| 29 | | | | 13–14 |
| 30 | | | | 14–15 |
| 31 | 15 | | | |
| 32 | 16 | | | |
| 36 | 3–8 | | | 9–16 |
| 37 | 4–8, 11–16 | | | 9–10 |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 38 | 5–8, 12–16 | | | 9–11 |
| 39 | 6–8, 13–16 | | | 9–12 |
| 40 | 7–8, 14–16 | | | 9–13 |
| 41 | 8, 12–16 | | | 9–11 |
| 42 | 13–16 | | | 9–12 |
| 43 | 11–16 | | | 10 |
| 44 | 12–16 | | | 11 |
| 45 | 13–16 | | | 12 |
| 46 | 14–16 | | | 13 |
| 47 | 14–16 | | | |
| 48 | 15–16 | | | |
| 49 | 16 | | | |
| 53 | 3–11, 15, 20–25 | | | 12–14, 16, 26 |
| 54 | 4–11, 21–25 | | | 12–16, 26–27 |
| 55 | 5–11, 22–25 | | | 12–16, 26–27 |
| 56 | 6–11, 23–25 | | | 12–16, 26–28 |
| 57 | 7–11, 24–25 | | | 12–16, 26–28 |
| 58 | 8–11, 25 | | | 12–16, 26–29 |
| 59 | 9–16, 26–28 | | | 29 |
| 60 | 10–16, 27–28 | | | 29–30 |
| 61 | 11–16, 28 | | | 29–30 |
| 62 | 12–16 | | | 29–31 |
| 63 | 13–16 | | | 30–31 |
| 64 | 14–16 | | | 31–32 |
| 65 | 15–16 | | | 32 |
| 66 | 16 | | | 33 |
| 70 | 3–8, 11 | | | 9–10, 12–16, 20–33 |
| 71 | 4–9, 12–13, 23–24, 27–29, 32–33 | | | 10–11, 14–16, 21–22, 25–26, 30–31 |
| 72 | 5–9, 13, 24, 28–29, 33 | | | 10–12, 14–16, 22–23, 25–27, 30–32 |
| 73 | 6–9, 11, 14, 23, 26, 29 | | | 10, 12–13, 15–16, 24–25, 27–28, 30–33 |
| 74 | 7–10 | | | 11–16, 24–33 |
| 75 | 8–10 | | | 11–16, 25–33 |
| 76 | 9–11 | | | 12–16, 26–33 |
| 77 | 10–11 | | | 12–16, 27–33 |
| 78 | 11 | | | 12–16, 28–33 |
| 79 | 29, 32 | | | 12–16, 30–31, 33 |
| 80 | | | | 13–16, 30–33 |
| 81 | | | | 14–16, 31–33 |
| 82 | | | | 15–16, 32–33 |
| 83 | 16, 33 | | | |
| 87 | 3–8 | | | 9–16, 20–33, 37–43 |
| 88 | 4–8, 11 | | | 9–10, 12–16, 21–33, 38–44 |
| 89 | 5–8, 10, 14–15, 26–27, 30–31, 39, 42–43 | | | 9, 11–13, 16, 22–25, 28–29, 32–33, 40–41, 44 |
| 90 | 6–8, 15, 27, 31, 43 | | | 9–14, 16, 23–26, 28–30, 32–33, 40–42, 44–45 |
| 91 | 7–8 | | | 9–16, 24–33, 41–45 |
| 92 | 8 | | | 9–16, 25–33, 42–46 |
| 93 | | | | 9–16, 26–33, 43–46 |
| 94 | 11 | | | 10, 12–16, 27–33, 44–47 |
| 95 | | | | 11–16, 28–33, 45–47 |
| 96 | | | | 12–16, 29–33, 46–48 |
| 97 | 47 | | | 13–16, 30–33, 48 |
| 98 | | | | 14–16, 31–33, 48–49 |
| 99 | | | | 15–16, 32–33, 49 |
| 100 | | | | 16, 33, 50 |

## LS[17]$(3, k, v)$

61 of 693 admissible parameter sets exist, 632 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 20 | | | | 4–10 |
| 21 | | | | 5–10 |
| 22 | | | | 6–11 |
| 23 | | | | 7–11 |
| 24 | 10 | | | 8–9, 11–12 |
| 25 | | | | 9–12 |
| 26 | | | | 10–13 |
| 27 | | | | 11–13 |
| 28 | | | | 12–14 |
| 29 | | | | 13–14 |
| 30 | | | | 14–15 |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 31 | | | | 15 |
| 32 | 16 | | | |
| 37 | | | | 4–16 |
| 38 | 12–16 | | | 5–11 |
| 39 | 13–16 | | | 6–12 |
| 40 | 14–16 | | | 7–13 |
| 41 | 15–16 | | | 8–14 |
| 42 | 13, 16 | | | 9–12, 14–15 |
| 43 | | | | 10–16 |
| 44 | 12–13 | | | 11, 14–16 |
| 45 | 13 | | | 12, 14–16 |
| 46 | | | | 13–16 |
| 47 | | | | 14–16 |
| 48 | 15 | | | 16 |
| 49 | | | | 16 |
| 54 | 4, 7–8, 10–11, 21, 23–25 | | | 5–6, 9, 12–16, 22, 26–27 |
| 55 | 8, 11, 24–25 | | | 5–7, 9–10, 12–16, 22–23, 26–27 |
| 56 | 25 | | | 6–16, 23–24, 26–28 |
| 57 | | | | 7–16, 24–28 |
| 58 | | | | 8–16, 25–29 |
| 59 | | | | 9–16, 26–29 |
| 60 | 11–16, 27 | | | 10, 28–30 |
| 61 | 12–16 | | | 11, 28–30 |
| 62 | 13–16 | | | 12, 29–31 |
| 63 | 14–16 | | | 13, 30–31 |
| 64 | 15–16 | | | 14, 31–32 |
| 65 | 16 | | | 15, 32 |
| 66 | | | | 16, 33 |
| 71 | | **4** | | 5–16, 21–33 |
| 72 | 13, 29 | | | 5–12, 14–16, 22–28, 30–33 |
| 73 | | | | 6–16, 23–33 |
| 74 | | | | 7–16, 24–33 |
| 75 | | | | 8–16, 25–33 |
| 76 | | | | 9–16, 26–33 |
| 77 | | | | 10–16, 27–33 |
| 78 | | | | 11–16, 28–33 |
| 79 | | | | 12–16, 29–33 |
| 80 | | | | 13–16, 30–33 |
| 81 | | | | 14–16, 31–33 |
| 82 | | | | 15–16, 32–33 |
| 83 | | | | 16, 33 |
| 88 | | | | 4–16, 21–33, 38–44 |
| 89 | | | | 5–16, 22–33, 39–44 |
| 90 | | | | 6–16, 23–33, 40–45 |
| 91 | | | | 7–16, 24–33, 41–45 |
| 92 | | | | 8–16, 25–33, 42–46 |
| 93 | | | | 9–16, 26–33, 43–46 |
| 94 | | | | 10–16, 27–33, 44–47 |
| 95 | | | | 11–16, 28–33, 45–47 |
| 96 | | | | 12–16, 29–33, 46–48 |
| 97 | | | | 13–16, 30–33, 47–48 |
| 98 | | | | 14–16, 31–33, 48–49 |
| 99 | | | | 15–16, 32–33, 49 |
| 100 | | | | 16, 33, 50 |

## LS[17](4, $k$, $v$)

None of the 594 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[17](5, $k$, $v$)

None of the 504 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[17](6, $k$, $v$)

None of the 420 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[19](2, $k$, $v$)

488 of 834 admissible parameter sets exist, 346 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 21 | 3–4 | | | 5–10 |
| 22 | 4 | **5** | | 6–11 |
| 23 | 5–10 | | | 11 |
| 24 | 6–10 | | | 11–12 |
| 25 | 7–11 | | | 12 |
| 26 | 8–11 | | | 12–13 |
| 27 | 9–11 | | | 12–13 |
| 28 | 10–11 | | | 12–14 |
| 29 | 11 | | | 12–14 |
| 30 | | | | 12–15 |
| 31 | 13–15 | | | |
| 32 | 14–16 | | | |
| 33 | 15–16 | | | |
| 34 | 16–17 | | | |
| 35 | 17 | | | |
| 36 | 18 | | | |
| 40 | 3–4 | | | 5–18 |
| 41 | 4, 12–14, 18 | | | 5–11, 15–17 |
| 42 | 13–14 | | | 5–12, 15–18 |
| 43 | 9–14, 16–18 | | | 6–8, 15 |
| 44 | 10–14, 17–18 | | | 7–9, 15–16 |
| 45 | 11–14, 18 | | | 8–10, 15–17 |
| 46 | 12–14 | | | 9–11, 15–18 |
| 47 | 10–18 | | | |
| 48 | 11–18 | | | |
| 49 | 12–18 | | | |
| 50 | 13–18 | | | |
| 51 | 14–18 | | | |
| 52 | 15–18 | | | |
| 53 | 16–18 | | | |
| 54 | 17–18 | | | |
| 55 | 18 | | | |
| 59 | 3–18, 22–28 | | | 29 |
| 60 | 4–18, 23–28 | | | 29–30 |
| 61 | 5–18, 24–28 | | | 29–30 |
| 62 | 6–18, 25–28 | | | 29–31 |
| 63 | 7–18, 26–28 | | | 29–31 |
| 64 | 8–18, 27–29 | | | 30–32 |
| 65 | 9–18, 28–29 | | | 30–32 |
| 66 | 10–18, 29 | | | 30–33 |
| 67 | 11–18, 32 | | | 30–31, 33 |
| 68 | 12–18 | | | 31–34 |
| 69 | 13–18 | | | 32–34 |
| 70 | 14–18 | | | 33–35 |
| 71 | 15–18, 34 | | | 35 |
| 72 | 16–18 | | | 35–36 |
| 73 | 17–18 | | | 36 |
| 74 | 18 | | | 37 |
| 78 | 3–4, 17–18 | | | 5–16, 22–37 |
| 79 | 4–5, 8, 11, 17–18, 23, 29, 32, 35 | | | 6–7, 9–10, 12–16, 24–28, 30–31, 33–34, 36–37 |
| 80 | 5–10, 13–18 | | | 11–12, 24–37 |
| 81 | 6–10, 14–18, 34 | | | 11–13, 25–33, 35–37 |
| 82 | 7–11, 14–18 | | | 12–13, 26–37 |
| 83 | 8–18, 27–37 | | | |
| 84 | 9–18, 28–37 | | | |
| 85 | 10–18, 29–37 | | | |
| 86 | 11–18, 30–37 | | | |
| 87 | 12–18, 31–37 | | | |
| 88 | 13–18, 32–37 | | | |
| 89 | 14–18, 33–37 | | | |
| 90 | 15–18, 34–37 | | | |
| 91 | 16–18, 35–37 | | | |
| 92 | 17–18, 36–37 | | | |
| 93 | 18, 37 | | | |
| 97 | 3–4, 35, 47 | | | 5–18, 22–34, 36–37, 41–46, 48 |
| 98 | 4, 12–14, 18 | | | 5–11, 15–17, 23–37, 42–49 |
| 99 | 13–14 | | | 5–12, 15–18, 24–37, 43–49 |
| 100 | 9–14, 16–18 | | | 6–8, 15, 25–37, 44–50 |

## LS[19]$(3, k, v)$

87 of 714 admissible parameter sets exist, 627 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 22 | | | | 4–11 |
| 23 | | **5** | | 6–11 |
| 24 | 7, 10 | | | 6, 8–9, 11–12 |
| 25 | | | | 7–12 |
| 26 | | | | 8–13 |
| 27 | | | | 9–13 |
| 28 | 11 | | | 10, 12–14 |
| 29 | | | | 11–14 |
| 30 | | | | 12–15 |
| 31 | | | | 13–15 |
| 32 | 14–16 | | | |
| 33 | 15–16 | | | |
| 34 | 16–17 | | | |
| 35 | 17 | | | |
| 36 | 18 | | | |
| 41 | | | | 4–18 |
| 42 | 13 | | | 5–12, 14–18 |
| 43 | | | | 6–18 |
| 44 | 10, 12–13, 17–18 | | | 7–9, 11, 14–16 |
| 45 | 13, 18 | | | 8–12, 14–17 |
| 46 | | | | 9–18 |
| 47 | | | | 10–18 |
| 48 | 11–15, 17–18 | | | 16 |
| 49 | 12–15, 18 | | | 16–17 |
| 50 | 13–15 | | | 16–18 |
| 51 | 14–15 | | | 16–18 |
| 52 | 15 | | | 16–18 |
| 53 | | | | 16–18 |
| 54 | 17 | | | 18 |
| 55 | | | | 18 |
| 60 | 4, 7–9, 11–17, 23, 27 | | | 5–6, 10, 18, 24–26, 28–30 |
| 61 | 8–9, 12–17 | | | 5–7, 10–11, 18, 24–30 |
| 62 | 9, 13–17 | | | 6–8, 10–12, 18, 25–31 |
| 63 | 14–17 | | | 7–13, 18, 26–31 |
| 64 | 15–17 | | | 8–14, 18, 27–32 |
| 65 | 16–17 | | | 9–15, 18, 28–32 |
| 66 | 17 | | | 10–16, 18, 29–33 |
| 67 | | | | 11–18, 30–33 |
| 68 | 15 | | | 12–14, 16–18, 31–34 |
| 69 | | | | 13–18, 32–34 |
| 70 | | | | 14–18, 33–35 |
| 71 | | | | 15–18, 34–35 |
| 72 | | | | 16–18, 35–36 |
| 73 | | | | 17–18, 36 |
| 74 | | | | 18, 37 |
| 79 | | | | 4–18, 23–37 |
| 80 | | | | 5–18, 24–37 |
| 81 | | | | 6–18, 25–37 |
| 82 | | | | 7–18, 26–37 |
| 83 | | | | 8–18, 27–37 |
| 84 | 9, 11, 13, 15, 17, 29, 31, 33, 37 | | | 10, 12, 14, 16, 18, 28, 30, 32, 34–36 |
| 85 | | | | 10–18, 29–37 |
| 86 | | | | 11–18, 30–37 |
| 87 | | | | 12–18, 31–37 |
| 88 | | | | 13–18, 32–37 |
| 89 | | | | 14–18, 33–37 |
| 90 | | | | 15–18, 34–37 |
| 91 | | | | 16–18, 35–37 |
| 92 | | | | 17–18, 36–37 |
| 93 | | | | 18, 37 |
| 98 | | | | 4–18, 23–37, 42–49 |
| 99 | | | | 5–18, 24–37, 43–49 |
| 100 | | | | 6–18, 25–37, 44–50 |

## LS[19]$(4, k, v)$

None of the 600 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[19]$(5, k, v)$

None of the 495 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[19](6, $k$, $v$)

None of the 396 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[23](2, $k$, $v$)

244 of 878 admissible parameter sets exist, 634 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 25 | 3–4, 11 | | | 5–10, 12 |
| 26 | 4 | | | 5–13 |
| 27 | 5, 7–8, 10–11 | | | 6, 9, 12–13 |
| 28 | 8, 11 | | | 6–7, 9–10, 12–14 |
| 29 | 10–11 | | | 7–9, 12–14 |
| 30 | 11 | | | 8–10, 12–15 |
| 31 | 9–15 | | | |
| 32 | 10–16 | | | |
| 33 | 11–16 | | | |
| 34 | 12–17 | | | |
| 35 | 13–17 | | | |
| 36 | 14–18 | | | |
| 37 | 15–18 | | | |
| 38 | 16–19 | | | |
| 39 | 17–19 | | | |
| 40 | 18–20 | | | |
| 41 | 19–20 | | | |
| 42 | 20–21 | | | |
| 43 | 21 | | | |
| 44 | 22 | | | |
| 48 | 3–4 | | | 5–22 |
| 49 | 4, 18–19 | | | 5–17, 20–22 |
| 50 | 19 | | | 5–18, 20–22 |
| 51 | | | | 6–22 |
| 52 | | | | 7–22 |
| 53 | 9–11, 15–22 | | | 8, 12–14 |
| 54 | 10–11, 16–22 | | | 9, 12–15 |
| 55 | 11, 17–22 | | | 10, 12–16 |
| 56 | 18–22 | | | 11–17 |
| 57 | 19–22 | | | 12–18 |
| 58 | 20–22 | | | 13–19 |
| 59 | 14–22 | | | |
| 60 | 15–22 | | | |
| 61 | 16–22 | | | |
| 62 | 17–22 | | | |
| 63 | 18–22 | | | |
| 64 | 19–22 | | | |
| 65 | 20–22 | | | |
| 66 | 21–22 | | | |
| 67 | 22 | | | |
| 71 | 3–4, 9, 12–13, 17–19, 27–29, 32–34 | | | 5–8, 10–11, 14–16, 20–22, 26, 30–31, 35 |
| 72 | 4, 13, 18–19, 28–29, 33–34 | | | 5–12, 14–17, 20–22, 27, 30–32, 35–36 |
| 73 | 11, 14, 19, 29, 34–35 | | | 5–10, 12–13, 15–18, 20–22, 28, 30–33, 36 |
| 74 | 35 | | | 6–22, 29–34, 36–37 |
| 75 | | | | 7–22, 30–37 |
| 76 | | | | 8–22, 31–38 |
| 77 | | | | 9–22, 32–38 |
| 78 | | | | 10–22, 33–39 |
| 79 | 11, 17, 20, 35, 38 | | | 12–16, 18–19, 21–22, 34, 36–37, 39 |
| 80 | | | | 12–22, 35–40 |
| 81 | 19, 22, 38 | | | 13–18, 20–21, 36–37, 39–40 |
| 82 | | | | 14–22, 37–41 |
| 83 | 15–22, 38–40 | | | 41 |
| 84 | 16–22, 39–40 | | | 41–42 |
| 85 | 17–22, 40 | | | 41–42 |
| 86 | 18–22 | | | 41–43 |
| 87 | 19–22 | | | 42–43 |
| 88 | 20–22 | | | 43–44 |
| 89 | 21–22 | | | 44 |
| 90 | 22 | | | 45 |
| 94 | 3–4 | | | 5–22, 26–45 |
| 95 | 4 | | | 5–22, 27–45 |
| 96 | | | | 5–22, 28–45 |
| 97 | 35, 38 | | | 6–22, 29–34, 36–37, 39–45 |
| 98 | | | | 7–22, 30–45 |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 99 | | | | 8–22, 31–45 |
| 100 | | | | 9–22, 32–45 |

## LS[23]$(3, k, v)$

81 of 778 admissible parameter sets exist, 697 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 26 | | | | 4–13 |
| 27 | | | | 5–13 |
| 28 | 11 | | | 6–10, 12–14 |
| 29 | | | | 7–14 |
| 30 | 11 | | | 8–10, 12–15 |
| 31 | | | | 9–15 |
| 32 | 10–16 | | | |
| 33 | 11–16 | | | |
| 34 | 12–17 | | | |
| 35 | 13–17 | | | |
| 36 | 14–18 | | | |
| 37 | 15–18 | | | |
| 38 | 16–19 | | | |
| 39 | 17–19 | | | |
| 40 | 18–20 | | | |
| 41 | 19–20 | | | |
| 42 | 20–21 | | | |
| 43 | 21 | | | |
| 44 | 22 | | | |
| 49 | | | | 4–22 |
| 50 | 19 | | | 5–18, 20–22 |
| 51 | | | | 6–22 |
| 52 | | | | 7–22 |
| 53 | | | | 8–22 |
| 54 | 10–11, 17, 19, 21 | | | 9, 12–16, 18, 20, 22 |
| 55 | 11 | | | 10, 12–22 |
| 56 | | | | 11–22 |
| 57 | | | | 12–22 |
| 58 | | | | 13–22 |
| 59 | | | | 14–22 |
| 60 | 15–17, 19–22 | | | 18 |
| 61 | 16–17, 20–22 | | | 18–19 |
| 62 | 17, 21–22 | | | 18–20 |
| 63 | 22 | | | 18–21 |
| 64 | | | | 19–22 |
| 65 | | | | 20–22 |
| 66 | | | | 21–22 |
| 67 | | | | 22 |
| 72 | 4, 13, 19, 29 | | | 5–12, 14–18, 20–22, 27–28, 30–36 |
| 73 | | | | 5–22, 28–36 |
| 74 | | | | 6–22, 29–37 |
| 75 | | | | 7–22, 30–37 |
| 76 | | | | 8–22, 31–38 |
| 77 | | | | 9–22, 32–38 |
| 78 | | | | 10–22, 33–39 |
| 79 | | | | 11–22, 34–39 |
| 80 | | | | 12–22, 35–40 |
| 81 | | | | 13–22, 36–40 |
| 82 | | | | 14–22, 37–41 |
| 83 | | | | 15–22, 38–41 |
| 84 | 17, 19, 39 | | | 16, 18, 20–22, 40–42 |
| 85 | | | | 17–22, 40–42 |
| 86 | | | | 18–22, 41–43 |
| 87 | | | | 19–22, 42–43 |
| 88 | | | | 20–22, 43–44 |
| 89 | | | | 21–22, 44 |
| 90 | | | | 22, 45 |
| 95 | | | | 4–22, 27–45 |
| 96 | | | | 5–22, 28–45 |
| 97 | | | | 6–22, 29–45 |
| 98 | | | | 7–22, 30–45 |
| 99 | | | | 8–22, 31–45 |
| 100 | | | | 9–22, 32–45 |

## LS[23](4, $k$, $v$)

None of the 682 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[23](5, $k$, $v$)

None of the 592 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[23](6, $k$, $v$)

None of the 506 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[29](2, $k$, $v$)

All 905 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ are realised.

## LS[29](3, $k$, $v$)

124 of 827 admissible parameter sets exist, 703 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 32 | 5–6, 9–16 | | | 4, 7–8 |
| 33 | 6, 10–16 | | | 5, 7–9 |
| 34 | 11–17 | | | 6–10 |
| 35 | 12–17 | | | 7–11 |
| 36 | 13–18 | | | 8–12 |
| 37 | 14–18 | | | 9–13 |
| 38 | 12–13, 15–19 | | | 10–11, 14 |
| 39 | 13, 16–19 | | | 11–12, 14–15 |
| 40 | 17–20 | | | 12–16 |
| 41 | 18–20 | | | 13–17 |
| 42 | 19–21 | | | 14–18 |
| 43 | 20–21 | | | 15–19 |
| 44 | 17–22 | | | 16 |
| 45 | 18–22 | | | 17 |
| 46 | 19–23 | | | 18 |
| 47 | 20–23 | | | 19 |
| 48 | 20–24 | | | |
| 49 | 21–24 | | | |
| 50 | 22–25 | | | |
| 51 | 23–25 | | | |
| 52 | 24–26 | | | |
| 53 | 25–26 | | | |
| 54 | 26–27 | | | |
| 55 | 27 | | | |
| 56 | 28 | | | |
| 61 | | | | 4–28 |
| 62 | | | | 5–28 |
| 63 | | | | 6–28 |
| 64 | | | | 7–28 |
| 65 | | | | 8–28 |
| 66 | | | | 9–28 |
| 67 | | | | 10–28 |
| 68 | 15, 19, 21, 25 | | | 11–14, 16–18, 20, 22–24, 26–28 |
| 69 | | | | 12–28 |
| 70 | | | | 13–28 |
| 71 | | | | 14–28 |
| 72 | 19 | | | 15–18, 20–28 |
| 73 | | | | 16–28 |
| 74 | | | | 17–28 |
| 75 | | | | 18–28 |
| 76 | | | | 19–28 |
| 77 | | | | 20–28 |
| 78 | | | | 21–28 |
| 79 | | | | 22–28 |
| 80 | | | | 23–28 |
| 81 | | | | 24–28 |
| 82 | | | | 25–28 |
| 83 | | | | 26–28 |
| 84 | 27 | | | 28 |
| 85 | | | | 28 |
| 90 | 4, 45 | | | 5–28, 33–44 |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 91 | | | | 5–28, 34–45 |
| 92 | 46 | | | 6–28, 35–45 |
| 93 | | | | 7–28, 36–46 |
| 94 | 47 | | | 8–28, 37–46 |
| 95 | | | | 9–28, 38–47 |
| 96 | 48 | | | 10–28, 39–47 |
| 97 | | | | 11–28, 40–48 |
| 98 | 49 | | | 12–28, 41–48 |
| 99 | | | | 13–28, 42–49 |
| 100 | 50 | | | 14–28, 43–49 |

## LS[29]$(4, k, v)$

1 of 751 admissible parameter sets exist, 750 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 33 | 10 | | | 5–9, 11–16 |
| 34 | | | | 6–17 |
| 35 | | | | 7–17 |
| 36 | | | | 8–18 |
| 37 | | | | 9–18 |
| 38 | | | | 10–19 |
| 39 | | | | 11–19 |
| 40 | | | | 12–20 |
| 41 | | | | 13–20 |
| 42 | | | | 14–21 |
| 43 | | | | 15–21 |
| 44 | | | | 16–22 |
| 45 | | | | 17–22 |
| 46 | | | | 18–23 |
| 47 | | | | 19–23 |
| 48 | | | | 20–24 |
| 49 | | | | 21–24 |
| 50 | | | | 22–25 |
| 51 | | | | 23–25 |
| 52 | | | | 24–26 |
| 53 | | | | 25–26 |
| 54 | | | | 26–27 |
| 55 | | | | 27 |
| 56 | | | | 28 |
| 62 | | | | 5–28 |
| 63 | | | | 6–28 |
| 64 | | | | 7–28 |
| 65 | | | | 8–28 |
| 66 | | | | 9–28 |
| 67 | | | | 10–28 |
| 68 | | | | 11–28 |
| 69 | | | | 12–28 |
| 70 | | | | 13–28 |
| 71 | | | | 14–28 |
| 72 | | | | 15–28 |
| 73 | | | | 16–28 |
| 74 | | | | 17–28 |
| 75 | | | | 18–28 |
| 76 | | | | 19–28 |
| 77 | | | | 20–28 |
| 78 | | | | 21–28 |
| 79 | | | | 22–28 |
| 80 | | | | 23–28 |
| 81 | | | | 24–28 |
| 82 | | | | 25–28 |
| 83 | | | | 26–28 |
| 84 | | | | 27–28 |
| 85 | | | | 28 |
| 91 | | | | 5–28, 34–45 |
| 92 | | | | 6–28, 35–46 |
| 93 | | | | 7–28, 36–46 |
| 94 | | | | 8–28, 37–47 |
| 95 | | | | 9–28, 38–47 |
| 96 | | | | 10–28, 39–48 |
| 97 | | | | 11–28, 40–48 |
| 98 | | | | 12–28, 41–49 |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 99 | | | | 13–28, 42–49 |
| 100 | | | | 14–28, 43–50 |

## LS[29]$(5, k, v)$

None of the 679 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[29]$(6, k, v)$

None of the 609 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[31]$(2, k, v)$

268 of 847 admissible parameter sets exist, 579 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 33 | 3 | | | 4–16 |
| 34 | | | | 4–17 |
| 35 | | | | 5–17 |
| 36 | | | | 6–18 |
| 37 | 11–18 | | | 7–10 |
| 38 | 12–19 | | | 8–11 |
| 39 | 13–19 | | | 9–12 |
| 40 | 14–20 | | | 10–13 |
| 41 | 12–20 | | | 11 |
| 42 | 13–21 | | | 12 |
| 43 | 14–21 | | | 13 |
| 44 | 15–22 | | | 14 |
| 45 | 16–22 | | | 15 |
| 46 | 17–23 | | | 16 |
| 47 | 17–23 | | | |
| 48 | 18–24 | | | |
| 49 | 19–24 | | | |
| 50 | 20–25 | | | |
| 51 | 21–25 | | | |
| 52 | 22–26 | | | |
| 53 | 23–26 | | | |
| 54 | 24–27 | | | |
| 55 | 25–27 | | | |
| 56 | 26–28 | | | |
| 57 | 27–28 | | | |
| 58 | 28–29 | | | |
| 59 | 29 | | | |
| 60 | 30 | | | |
| 64 | 3, 5, 11, 13, 17, 23, 25, 29 | | | 4, 6–10, 12, 14–16, 18–22, 24, 26–28, 30 |
| 65 | | | | 4–30 |
| 66 | | | | 5–30 |
| 67 | 6–8, 14–21, 24–26, 29 | | | 9–13, 22–23, 27–28, 30 |
| 68 | 7–8, 15–21, 25–26 | | | 9–14, 22–24, 27–30 |
| 69 | 8, 16–21, 26 | | | 9–15, 22–25, 27–30 |
| 70 | 17–21 | | | 9–16, 22–30 |
| 71 | 12–13, 17–21, 23–24, 27–29 | | | 10–11, 14–16, 22, 25–26, 30 |
| 72 | 13, 18–21, 24, 28–29 | | | 11–12, 14–17, 22–23, 25–27, 30 |
| 73 | 14, 19–21, 23, 26, 29 | | | 12–13, 15–18, 22, 24–25, 27–28, 30 |
| 74 | 20–21 | | | 13–19, 22–30 |
| 75 | 21 | | | 14–20, 22–30 |
| 76 | | | | 15–30 |
| 77 | | | | 16–30 |
| 78 | | | | 17–30 |
| 79 | 20, 23, 29 | | | 18–19, 21–22, 24–28, 30 |
| 80 | | | | 19–30 |
| 81 | 22–23 | | | 20–21, 24–30 |
| 82 | 23 | | | 21–22, 24–30 |
| 83 | 22–30 | | | |
| 84 | 23–30 | | | |
| 85 | 24–30 | | | |
| 86 | 25–30 | | | |
| 87 | 26–30 | | | |
| 88 | 27–30 | | | |
| 89 | 28–30 | | | |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 90 | 29–30 | | | |
| 91 | 30 | | | |
| 95 | 3–4 | | | 5–30, 34–47 |
| 96 | 4 | | | 5–30, 35–48 |
| 97 | 38, 47 | | | 5–30, 36–37, 39–46, 48 |
| 98 | | | | 6–30, 37–49 |
| 99 | | | | 7–30, 38–49 |
| 100 | | | | 8–30, 39–50 |

## LS[31]$(3, k, v)$

107 of 763 admissible parameter sets exist, 656 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 34 | | | | 4–17 |
| 35 | | | | 5–17 |
| 36 | | | | 6–18 |
| 37 | | | | 7–18 |
| 38 | 12–13, 15–19 | | | 8–11, 14 |
| 39 | 13, 16–19 | | | 9–12, 14–15 |
| 40 | 17–20 | | | 10–16 |
| 41 | 18–20 | | | 11–17 |
| 42 | 13, 19–21 | | | 12, 14–18 |
| 43 | 20–21 | | | 13–19 |
| 44 | 17–22 | | | 14–16 |
| 45 | 18–22 | | | 15–17 |
| 46 | 19–23 | | | 16–18 |
| 47 | 20–23 | | | 17–19 |
| 48 | 18–24 | | | |
| 49 | 19–24 | | | |
| 50 | 20–25 | | | |
| 51 | 21–25 | | | |
| 52 | 22–26 | | | |
| 53 | 23–26 | | | |
| 54 | 24–27 | | | |
| 55 | 25–27 | | | |
| 56 | 26–28 | | | |
| 57 | 27–28 | | | |
| 58 | 28–29 | | | |
| 59 | 29 | | | |
| 60 | 30 | | | |
| 65 | | | | 4–30 |
| 66 | | | | 5–30 |
| 67 | | | | 6–30 |
| 68 | 7, 15, 19, 21, 25 | | | 8–14, 16–18, 20, 22–24, 26–30 |
| 69 | | | | 8–30 |
| 70 | | | | 9–30 |
| 71 | | | | 10–30 |
| 72 | 13, 19, 29 | | | 11–12, 14–18, 20–28, 30 |
| 73 | | | | 12–30 |
| 74 | | | | 13–30 |
| 75 | | | | 14–30 |
| 76 | | | | 15–30 |
| 77 | | | | 16–30 |
| 78 | | | | 17–30 |
| 79 | | | | 18–30 |
| 80 | | | | 19–30 |
| 81 | | | | 20–30 |
| 82 | | | | 21–30 |
| 83 | | | | 22–30 |
| 84 | 23, 25, 27, 29 | | | 24, 26, 28, 30 |
| 85 | | | | 24–30 |
| 86 | | | | 25–30 |
| 87 | | | | 26–30 |
| 88 | | | | 27–30 |
| 89 | | | | 28–30 |
| 90 | | | | 29–30 |
| 91 | | | | 30 |
| 96 | 4 | | | 5–30, 35–48 |
| 97 | | | | 5–30, 36–48 |
| 98 | | | | 6–30, 37–49 |
| 99 | | | | 7–30, 38–49 |

| $v$ | exists, $k=$ | new | none | open, $k=$ |
|---|---|---|---|---|
| 100 | | | | 8–30, 39–50 |

## LS[31]$(4,k,v)$

None of the 681 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[31]$(5,k,v)$

None of the 603 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[31]$(6,k,v)$

None of the 527 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[37]$(2,k,v)$

475 of 856 admissible parameter sets exist, 381 are open.

| $v$ | exists, $k=$ | new | none | open, $k=$ |
|---|---|---|---|---|
| 39 | 3 | | | 4–19 |
| 40 | | | | 4–20 |
| 41 | 14, 18–19 | | | 5–13, 15–17, 20 |
| 42 | 19 | | | 6–18, 20–21 |
| 43 | 11, 17–20 | | | 7–10, 12–16, 21 |
| 44 | 18–20 | | | 8–17, 21–22 |
| 45 | 19–20 | | | 9–18, 21–22 |
| 46 | 20 | | | 10–19, 21–23 |
| 47 | 11–22 | | | 23 |
| 48 | 12–22 | | | 23–24 |
| 49 | 13–23 | | | 24 |
| 50 | 14–23 | | | 24–25 |
| 51 | 15–23 | | | 24–25 |
| 52 | 16–23 | | | 24–26 |
| 53 | 17–23 | | | 24–26 |
| 54 | 18–23 | | | 24–27 |
| 55 | 19–23 | | | 24–27 |
| 56 | 20–23 | | | 24–28 |
| 57 | 21–23 | | | 24–28 |
| 58 | 22–23 | | | 24–29 |
| 59 | 23–28 | | | 29 |
| 60 | 24–28 | | | 29–30 |
| 61 | 25–28 | | | 29–30 |
| 62 | 26–28 | | | 29–31 |
| 63 | 27–28 | | | 29–31 |
| 64 | 28–29 | | | 30–32 |
| 65 | 29 | | | 30–32 |
| 66 | | | | 30–33 |
| 67 | 32 | | | 31, 33 |
| 68 | | | | 32–34 |
| 69 | | | | 33–34 |
| 70 | | | | 34–35 |
| 71 | | | | 35 |
| 72 | | | | 36 |
| 76 | 3 | | | 4–36 |
| 77 | | | | 4–36 |
| 78 | | | | 5–36 |
| 79 | 8, 11, 17, 20, 23, 29, 32, 35 | | | 6–7, 9–10, 12–16, 18–19, 21–22, 24–28, 30–31, 33–34, 36 |
| 80 | | | | 7–36 |
| 81 | 19, 22–23, 34 | | | 8–18, 20–21, 24–33, 35–36 |
| 82 | 23 | | | 9–22, 24–36 |
| 83 | 10–36 | | | |
| 84 | 11–36 | | | |
| 85 | 12–36 | | | |
| 86 | 13–36 | | | |
| 87 | 14–36 | | | |
| 88 | 15–36 | | | |
| 89 | 16–36 | | | |
| 90 | 17–36 | | | |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 91 | 18–36 | | | |
| 92 | 19–36 | | | |
| 93 | 20–36 | | | |
| 94 | 21–36 | | | |
| 95 | 22–36 | | | |
| 96 | 23–36 | | | |
| 97 | 24–36 | | | |
| 98 | 25–36 | | | |
| 99 | 26–36 | | | |
| 100 | 27–36 | | | |

## LS[37]$(3, k, v)$

16 of 805 admissible parameter sets exist, 789 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 40 | | | | 4–20 |
| 41 | | | | 5–20 |
| 42 | | | | 6–21 |
| 43 | | | | 7–21 |
| 44 | 19 | | | 8–18, 20–22 |
| 45 | | | | 9–22 |
| 46 | | | | 10–23 |
| 47 | | | | 11–23 |
| 48 | 13, 15, 17, 19 | | | 12, 14, 16, 18, 20–24 |
| 49 | | | | 13–24 |
| 50 | | | | 14–25 |
| 51 | | | | 15–25 |
| 52 | | | | 16–26 |
| 53 | | | | 17–26 |
| 54 | | | | 18–27 |
| 55 | | | | 19–27 |
| 56 | | | | 20–28 |
| 57 | | | | 21–28 |
| 58 | | | | 22–29 |
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| 60 | | | | 24–30 |
| 61 | | | | 25–30 |
| 62 | | | | 26–31 |
| 63 | | | | 27–31 |
| 64 | | | | 28–32 |
| 65 | | | | 29–32 |
| 66 | | | | 30–33 |
| 67 | | | | 31–33 |
| 68 | | | | 32–34 |
| 69 | | | | 33–34 |
| 70 | | | | 34–35 |
| 71 | | | | 35 |
| 72 | | | | 36 |
| 77 | | | | 4–36 |
| 78 | | | | 5–36 |
| 79 | | | | 6–36 |
| 80 | | | | 7–36 |
| 81 | | | | 8–36 |
| 82 | | | | 9–36 |
| 83 | | | | 10–36 |
| 84 | 11, 13, 15, 17, 19, 23, 25, 27, 29, 31, 33 | | | 12, 14, 16, 18, 20–22, 24, 26, 28, 30, 32, 34–36 |
| 85 | | | | 12–36 |
| 86 | | | | 13–36 |
| 87 | | | | 14–36 |
| 88 | | | | 15–36 |
| 89 | | | | 16–36 |
| 90 | | | | 17–36 |
| 91 | | | | 18–36 |
| 92 | | | | 19–36 |
| 93 | | | | 20–36 |
| 94 | | | | 21–36 |
| 95 | | | | 22–36 |
| 96 | | | | 23–36 |
| 97 | | | | 24–36 |
| 98 | | | | 25–36 |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 99 | | | | 26–36 |
| 100 | | | | 27–36 |

## LS[37](4, $k$, $v$)

None of the 755 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[37](5, $k$, $v$)

None of the 707 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[37](6, $k$, $v$)

None of the 660 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[41](2, $k$, $v$)

257 of 890 admissible parameter sets exist, 633 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 43 | 3, 5, 11–13, 16–20 | | | 4, 6–10, 14–15, 21 |
| 44 | 12–13, 17–20 | | | 4–11, 14–16, 21–22 |
| 45 | 13, 18–20 | | | 5–12, 14–17, 21–22 |
| 46 | 19–20 | | | 6–18, 21–23 |
| 47 | 7–22 | | | 23 |
| 48 | 8–22 | | | 23–24 |
| 49 | 9–23 | | | 24 |
| 50 | 10–23 | | | 24–25 |
| 51 | 11–23 | | | 24–25 |
| 52 | 12–23 | | | 24–26 |
| 53 | 13–25 | | | 26 |
| 54 | 14–25 | | | 26–27 |
| 55 | 15–25 | | | 26–27 |
| 56 | 16–25 | | | 26–28 |
| 57 | 17–25 | | | 26–28 |
| 58 | 18–25 | | | 26–29 |
| 59 | 19–28 | | | 29 |
| 60 | 20–28 | | | 29–30 |
| 61 | 21–28 | | | 29–30 |
| 62 | 22–28 | | | 29–31 |
| 63 | 23–28 | | | 29–31 |
| 64 | 24–29 | | | 30–32 |
| 65 | 25–29 | | | 30–32 |
| 66 | 26–29 | | | 30–33 |
| 67 | 27–29, 32 | | | 30–31, 33 |
| 68 | 28–29 | | | 30–34 |
| 69 | 29 | | | 30–34 |
| 70 | | | | 30–35 |
| 71 | 32–34 | | | 31, 35 |
| 72 | 33–34 | | | 32, 35–36 |
| 73 | 34–35 | | | 33, 36 |
| 74 | 35 | | | 34, 36–37 |
| 75 | | | | 35–37 |
| 76 | | | | 36–38 |
| 77 | | | | 37–38 |
| 78 | | | | 38–39 |
| 79 | | | | 39 |
| 80 | | | | 40 |
| 84 | 3 | | | 4–40 |
| 85 | | | | 4–40 |
| 86 | | | | 5–40 |
| 87 | | | | 6–40 |
| 88 | | | | 7–40 |
| 89 | 10, 14–15, 26–27, 30–31, 35, 38–39 | | | 8–9, 11–13, 16–25, 28–29, 32–34, 36–37, 40 |
| 90 | 15, 27, 31, 39 | | | 9–14, 16–26, 28–30, 32–38, 40 |
| 91 | | | | 10–40 |
| 92 | | | | 11–40 |
| 93 | | | | 12–40 |
| 94 | | | | 13–40 |

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 95 | | | | 14–40 |
| 96 | | | | 15–40 |
| 97 | 35, 38 | | | 16–34, 36–37, 39–40 |
| 98 | | | | 17–40 |
| 99 | | | | 18–40 |
| 100 | | | | 19–40 |

## LS[41]$(3, k, v)$

74 of 833 admissible parameter sets exist, 759 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 44 | 13, 17–20 | | | 4–12, 14–16, 21–22 |
| 45 | 18–20 | | | 5–17, 21–22 |
| 46 | 19–20 | | | 6–18, 21–23 |
| 47 | 20 | | | 7–19, 21–23 |
| 48 | 10–11, 13–15, 17–22 | | | 8–9, 12, 16, 23–24 |
| 49 | 11, 14–15, 18–22 | | | 9–10, 12–13, 16–17, 23–24 |
| 50 | 15, 19–23 | | | 10–14, 16–18, 24–25 |
| 51 | 20–23 | | | 11–19, 24–25 |
| 52 | 21–23 | | | 12–20, 24–26 |
| 53 | 22–23 | | | 13–21, 24–26 |
| 54 | 17, 19, 21, 23–25 | | | 14–16, 18, 20, 22, 26–27 |
| 55 | 24–25 | | | 15–23, 26–27 |
| 56 | 25 | | | 16–24, 26–28 |
| 57 | | | | 17–28 |
| 58 | | | | 18–29 |
| 59 | | | | 19–29 |
| 60 | 20–23, 25–28 | | | 24, 29–30 |
| 61 | 21–23, 26–28 | | | 24–25, 29–30 |
| 62 | 22–23, 27–28 | | | 24–26, 29–31 |
| 63 | 23, 28 | | | 24–27, 29–31 |
| 64 | | | | 24–32 |
| 65 | | | | 25–32 |
| 66 | | | | 26–33 |
| 67 | | | | 27–33 |
| 68 | | | | 28–34 |
| 69 | | | | 29–34 |
| 70 | | | | 30–35 |
| 71 | | | | 31–35 |
| 72 | | | | 32–36 |
| 73 | | | | 33–36 |
| 74 | | | | 34–37 |
| 75 | | | | 35–37 |
| 76 | | | | 36–38 |
| 77 | | | | 37–38 |
| 78 | | | | 38–39 |
| 79 | | | | 39 |
| 80 | | | | 40 |
| 85 | | | | 4–40 |
| 86 | | | | 5–40 |
| 87 | | | | 6–40 |
| 88 | | | | 7–40 |
| 89 | | | | 8–40 |
| 90 | | | | 9–40 |
| 91 | | | | 10–40 |
| 92 | | | | 11–40 |
| 93 | | | | 12–40 |
| 94 | | | | 13–40 |
| 95 | | | | 14–40 |
| 96 | | | | 15–40 |
| 97 | | | | 16–40 |
| 98 | | | | 17–40 |
| 99 | | | | 18–40 |
| 100 | | | | 19–40 |

## LS[41]$(4, k, v)$

None of the 777 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[41](5, $k$, $v$)

None of the 723 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[41](6, $k$, $v$)

None of the 670 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[43](2, $k$, $v$)

287 of 862 admissible parameter sets exist, 575 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 45 | 3 | | | 4–22 |
| 46 | | | | 4–23 |
| 47 | 5–22 | | | 23 |
| 48 | 6–22 | | | 23–24 |
| 49 | 7–23 | | | 24 |
| 50 | 8–23 | | | 24–25 |
| 51 | 9–23 | | | 24–25 |
| 52 | 10–23 | | | 24–26 |
| 53 | 11–25 | | | 26 |
| 54 | 12–25 | | | 26–27 |
| 55 | 13–25 | | | 26–27 |
| 56 | 14–25 | | | 26–28 |
| 57 | 15–25 | | | 26–28 |
| 58 | 16–25 | | | 26–29 |
| 59 | 17–28 | | | 29 |
| 60 | 18–28 | | | 29–30 |
| 61 | 19–28 | | | 29–30 |
| 62 | 20–28 | | | 29–31 |
| 63 | 21–28 | | | 29–31 |
| 64 | 22–29 | | | 30–32 |
| 65 | 23–29 | | | 30–32 |
| 66 | 24–29 | | | 30–33 |
| 67 | 25–29, 32 | | | 30–31, 33 |
| 68 | 26–29 | | | 30–34 |
| 69 | 27–29 | | | 30–34 |
| 70 | 28–29 | | | 30–35 |
| 71 | 29, 32–34 | | | 30–31, 35 |
| 72 | 33–34 | | | 30–32, 35–36 |
| 73 | 34–35 | | | 31–33, 36 |
| 74 | 35 | | | 32–34, 36–37 |
| 75 | | | | 33–37 |
| 76 | | | | 34–38 |
| 77 | | | | 35–38 |
| 78 | | | | 36–39 |
| 79 | 38 | | | 37, 39 |
| 80 | | | | 38–40 |
| 81 | | | | 39–40 |
| 82 | | | | 40–41 |
| 83 | | | | 41 |
| 84 | | | | 42 |
| 88 | 3 | | | 4–42 |
| 89 | 10, 14–15, 26–27, 30–31, 35, 38–39, 42 | | | 4–9, 11–13, 16–25, 28–29, 32–34, 36–37, 40–41 |
| 90 | 15, 27, 31, 39 | | | 5–14, 16–26, 28–30, 32–38, 40–42 |
| 91 | | | | 6–42 |
| 92 | | | | 7–42 |
| 93 | | | | 8–42 |
| 94 | | | | 9–42 |
| 95 | | | | 10–42 |
| 96 | | | | 11–42 |
| 97 | 35, 38 | | | 12–34, 36–37, 39–42 |
| 98 | | | | 13–42 |
| 99 | | | | 14–42 |
| 100 | | | | 15–42 |

## LS[43](3, $k$, $v$)

60 of 802 admissible parameter sets exist, 742 are open.

| $v$ | exists, $k =$ | new | none | open, $k =$ |
|---|---|---|---|---|
| 46 | | | | 4–23 |
| 47 | | | | 5–23 |
| 48 | 6–7, 10–11, 13–15, 17–22 | | | 8–9, 12, 16, 23–24 |
| 49 | 7, 11, 14–15, 18–22 | | | 8–10, 12–13, 16–17, 23–24 |
| 50 | 15, 19–23 | | | 8–14, 16–18, 24–25 |
| 51 | 20–23 | | | 9–19, 24–25 |
| 52 | 21–23 | | | 10–20, 24–26 |
| 53 | 22–23 | | | 11–21, 24–26 |
| 54 | 17, 19, 21, 23, 25 | | | 12–16, 18, 20, 22, 24, 26–27 |
| 55 | | | | 13–27 |
| 56 | | | | 14–28 |
| 57 | | | | 15–28 |
| 58 | | | | 16–29 |
| 59 | | | | 17–29 |
| 60 | 19–23, 26–27 | | | 18, 24–25, 28–30 |
| 61 | 20–23, 27 | | | 19, 24–26, 28–30 |
| 62 | 21–23 | | | 20, 24–31 |
| 63 | 22–23 | | | 21, 24–31 |
| 64 | 23 | | | 22, 24–32 |
| 65 | | | | 23–32 |
| 66 | | | | 24–33 |
| 67 | | | | 25–33 |
| 68 | | | | 26–34 |
| 69 | | | | 27–34 |
| 70 | | | | 28–35 |
| 71 | | | | 29–35 |
| 72 | | | | 30–36 |
| 73 | | | | 31–36 |
| 74 | | | | 32–37 |
| 75 | | | | 33–37 |
| 76 | | | | 34–38 |
| 77 | | | | 35–38 |
| 78 | | | | 36–39 |
| 79 | | | | 37–39 |
| 80 | | | | 38–40 |
| 81 | | | | 39–40 |
| 82 | | | | 40–41 |
| 83 | | | | 41 |
| 84 | | | | 42 |
| 89 | | | | 4–42 |
| 90 | | | | 5–42 |
| 91 | | | | 6–42 |
| 92 | | | | 7–42 |
| 93 | | | | 8–42 |
| 94 | | | | 9–42 |
| 95 | | | | 10–42 |
| 96 | | | | 11–42 |
| 97 | | | | 12–42 |
| 98 | | | | 13–42 |
| 99 | | | | 14–42 |
| 100 | | | | 15–42 |

## LS[43](4, $k$, $v$)

None of the 743 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[43](5, $k$, $v$)

None of the 686 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.

## LS[43](6, $k$, $v$)

None of the 630 admissible parameter sets with $v \leq 100$ and $k \leq v/2$ is known to be realised.